Arithmetic Expression Geometry I: Foundations

Sequential and Branching Histories, Order-Topological Calculus, and Hole–Cap Geometry

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Draft notice. This manuscript is a working draft in the editorial and publication sense. Here "draft" is not a mathematical claim-status label and does not alter the status of any theorem, proposition, computation, conjecture, or open problem. The geometric chapters that earlier versions of this paper contained are now developed in Paper 0; the migration is recorded in `governance/00b-paper-0-geometric-foundation-amendment.md`.

Abstract

Ordinary arithmetic evaluation collapses an expression tree into an endpoint value and discards the order of the computation. This paper develops the syntax of that discarded order, its order-topological and analytic structure, and the zero and hole–cap geometry of the spaces on which the words act. A finite binary expression tree has a unique legal internal evaluation order exactly when its internal dependency poset is a chain, equivalently when its internal vertices form a single spine; marking an accumulator then makes fully marked planar trees and chronological words of one-hole contexts correspond bijectively. Operand-slot chirality is retained as explicit data, distinct from planar mirror, temporal reversal, and path inverse. For a general branching tree the dependency poset carries a down-set information order whose ideal completion is the dcpo of finite and (finitely predicated) infinite computation DAGs, and the partial evaluations of a fixed tree form a finite distributive lattice whose maximal chains are exactly the spinal histories (Birkhoff duality). On the continuous lattice $[0, \infty]$, the elementary contexts realize the Scott and Lawson structure: addition is the way-below witness, scaling is a Scott automorphism, and the reciprocal is the order-reversing involution whose adjoining turns the one-sided Scott topology into the two-sided Hausdorff Lawson topology, mirroring the passage from the positive affine semigroup to $\mathrm{PGL}_2(\mathbb{R})$. For non-degenerate bilateral contexts over an arbitrary field K, projective evaluation is compatible with chronological composition; the elementary contexts generate $\mathrm{PGL}_2(K)$, those fixing infinity form the affine sector $B_\infty \cong \mathrm{Aff}(1, K)$ onto which the

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affine sublanguage maps surjectively, and the same case division yields the rank-one Bruhat decomposition. Over $\mathbb{Q}(\sqrt{2})$, the three-context sublanguage generated by $z \mapsto z \pm \sqrt{2}$ and $z \mapsto -1/z$ realizes the $q = 4$ Hecke group, and the order-four relation $(JT)^4 = 1$ is carried by an explicit eight-step marked history that is projectively trivial yet ordinarily inadmissible at $x = -\sqrt{2}$; projective non-degeneracy is therefore kept separate from ordinary admissibility. The analysis of these histories has three pillars. The differential pillar identifies the order defect of the minimal branching DAG $\{\text{add}, \text{multiply}\}$ with the contact curvature $[D_u, D_v] = \mu\lambda\,\partial_a$. The integral pillar expresses the order defect as a group-valued 1-cocycle on the linear-extension graph, closed by the Jacobi identity and by the holographic Stokes theorem, which pushes the bulk curvature to the boundary holonomy. The series pillar expands the history evaluation into the free Lie algebra as a Magnus series, whose arity-k brackets are carried by the $(k-1)$-cells of the down-set lattice and whose abelianization is the difference substitution. On the geometric side, five notions of hole and four of boundary are kept distinct; a decorated network thickened to a ball-and-tube body exhibits the hole–disk complementarity "the surface gives the hole its meaning, the body gives the disk its meaning", and a constant-boundary disk admits no regular AES capping. In a regular arithmetic expression space, subject to the eikonal axiom $|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2$ with $\mu \neq 0$, the zero locus of the assignment is a smooth curve with no isolated point, crossing, or branch point; on a connected, complete, boundaryless space the unit-gradient rectification of the assignment gives a diffeomorphism $M \cong Z(a) \times \mathbb{R}$, so branching zero networks are excluded in the regular category. Singular arithmetic expression spaces, whose singular set is closed, nowhere dense, and locally essential, are then defined, with two essential puncture mechanisms (non-C^1 assignment versus conformal-background degeneration); a disc model with an isolated non-C^1 zero is verified, and parameter families with spatially regular zeros are shown to have submersive total zero sets under a stated properness hypothesis.

Keywords: arithmetic expression geometry; binary expression trees; dependency posets and linear extensions; one-hole contexts; down-set dcpo and Birkhoff duality; Scott–Lawson topology; projective linear group; Bruhat decomposition; Hecke groups; order defect and contact curvature; Magnus series and free Lie algebra; hole–cap geometry; eikonal equation and zero loci; complete splitting theorem; singular arithmetic expression spaces

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Interface: From Paper 0 to Paper I

Paper 0 is the geometric foundation of this series. It constructs the space in which the elementary arithmetic operations act as motions, the dual reading of those motions, the commutativization against which their order defect is measured, and the contact structure of that defect. It fixes explicit left- and right-expanded conventions, proves the distributive expansion that relates two expansions sharing an operator, and treats reciprocals, poles, the projective point at infinity, affine and reciprocal fixed points, and the common 2×2 matrix representation.

The present paper begins one level earlier and stays there. Its first problem is intrinsic: which finite binary expression trees have a unique legal order of internal evaluation, including mixed operand-slot patterns? The answer is the single-spine classification. Choosing an accumulator then gives the marked spinal histories used throughout this paper. We retain that terminology and formalism unchanged.

The division of labour is therefore the following.

- **Paper 0:** expansions, arithmetic motions, the models $\mathfrak{E}_0$ and $\mathfrak{E}_1$, ripple geometry, affine cocycles, the accumulative commutative space and tearing, and the contact structure of the tearing.
- **Paper I:** intrinsic sequential and branching histories, bilateral projective evaluation, the Scott–Lawson order topology of histories, the analysis three pillars (differential, integral, series), and the hole–cap fine geometry of the regular and singular zero theory.

Per the amendment recorded in `governance/00d-paper-I-calculus-and-capping-amendment.md`, the analysis and the hole–cap geometry are now developed *here*; Paper 0's contact, ACS, and punctured-AES chapters are the canonical upstream, and this paper restates the specific primitives it needs at the same labels.

This paper imports, with labels and proofs unchanged from Paper 0:

1. the regular arithmetic expression space $(M, g, a; \mu, \lambda)$ and its eikonal axiom $|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2$ with $\mu \neq 0$ (`def:regular-aes`), together with the canonical arithmetic frame (`prop:canonical-arithmetic-f`
2. the arithmetic contact model $\mathcal{C} = \mathbb{R}^3_{(u,v,a)}$, the contact one-form $\alpha = da - \mu\, du - \lambda a\, dv$ (`eq:arithmetic-contact-form`), the horizontal fields $D_u = \partial_u + \mu\partial_a$, $D_v = \partial_v + \lambda a \partial_a$ (`eq:horizontal-arithmetic-fields`), and the contact curvature bracket $[D_u, D_v] = \mu\lambda\, \partial_a$ (`thm:contact-curvature`);
3. the accumulative commutative space $\mathcal{A} = \mathbb{R}^2_{(A,M)}$ and the weighted torsion–Stokes theorem (`thm:torsion-stokes`);
4. the punctured AES (`def:p0-punctured-aes`) and the two-sided collar contract (`def:p0-collar-contract`).

These imports are restated in the chapters that use them; no theorem of Paper 0 is re-proved here, and the ordinary-admissibility / projective-continuation distinction remains active throughout.

1 Introduction

An arithmetic expression has several mathematically distinct lives. It is a finite syntax tree; it may be equipped with a legal order of evaluation; a chosen running value turns that order into a marked history; the history induces an operator; and the operator sends an initial value to an endpoint.

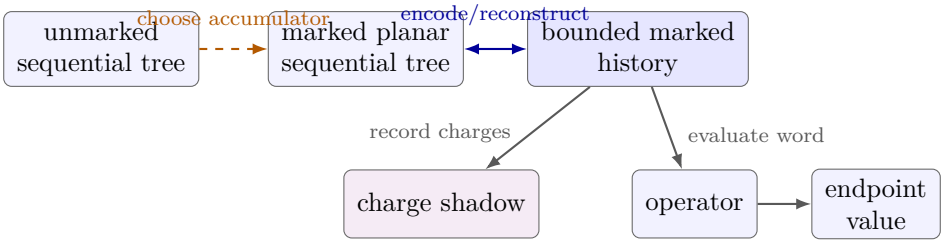


Figure 1: The process-to-result structure. Choosing an accumulator marks an unmarked sequential tree; a fully marked planar tree and a bounded marked history determine one another. From the history, operator evaluation and the charge map are separate information-losing branches, while an endpoint is obtained by applying the operator to an initial value.

Ordinary arithmetic usually passes through these levels as quickly as possible. AEG instead asks what is lost at each passage.

The guiding structure of this paper is

$$\begin{gathered}\text{unmarked sequential tree} \xrightarrow{\text{choose an accumulator}} \text{marked planar sequential tree},\\ \text{marked planar sequential tree} \longleftrightarrow \text{bounded marked history},\\ \text{bounded marked history} \longrightarrow \text{operator} \longrightarrow \text{endpoint value},\\ \text{bounded marked history} \longrightarrow \text{charge shadow}.\end{gathered} \tag{1}$$

The first arrow denotes a choice, not a canonical map. When the planar structure, mark, operation labels, constants, and slots are retained, the double arrow is the bijection proved in Section 2. The two arrows leaving a history are separate forgetful maps: charge is not a function of the evaluated operator, and an endpoint is not a function of charge alone. Different histories may nevertheless induce the same operator, and different operators may agree at one chosen initial value. The geometry belongs primarily to these intermediate levels, before endpoint evaluation has erased the order of construction.

Figure 1 renders this structure explicitly. It is a map of the distinctions used in the paper, not a claim that any forgetful arrow is injective.

1.1 Sequential histories

Not every expression tree determines a unique history. Independent subexpressions may be evaluated in either order, so a single tree can have many legal linear extensions of its dependency order. Section 2 isolates the precise exception. For a finite binary tree, uniqueness of the internal evaluation order is equivalent to the internal vertices forming one chain, or equivalently one spine. A marked accumulator on that spine determines a word of one-hole contexts

$$C^{(1)}_{\omega,c}[z] = \omega(z, c), \qquad C^{(2)}_{\omega,c}[z] = \omega(c, z).$$

The superscript records the operand slot occupied by the current value. It is not a time direction. This distinction prevents three operations that are often blurred in informal tree pictures—planar mirror, temporal reversal, and path inverse—from being identified.

This use of syntax is modest. Formal generation and rewriting have long histories in logic and theoretical computer science; classical points of reference include Post's work on combinatorial reduction and Chomsky's generative hierarchies [4, 1]. Here the purpose is different: the sequential fragment supplies a controlled domain on which evaluation can be treated as a path without pretending that every arithmetic tree has a unique path.

1.2 Why bilateral completion matters

Earlier affine calculations use the current value in one distinguished operand slot. That fragment is closed under translations and nonzero scalings and is therefore naturally represented by the affine group of the line. Once both operand slots are allowed, the context $z \mapsto c/z$ introduces inversion. The correct algebraic ambient space is then the projective line and its fractional linear transformations.

Section 3 records, over an arbitrary field K, that the non-degenerate bilateral contexts generate $\mathrm{PGL}_2(K)$; the generation theorem itself is proved in Paper 0, and what this paper adds is its reading through the history language. The affine transformations are exactly the stabilizer of the projective point at infinity. Thus the continuous geometry is neither discarded nor promoted to the whole theory: it is identified precisely as the affine sector of a larger projective semantics. The geometric reading of these operators — addition and multiplication as motions of an arithmetic expression space, and second-slot division as the projective step — is developed in Paper 0 and used here only at the operator level.

Inside that larger semantics, the two contexts

$$T(z) = z + \sqrt{2}, \qquad J(z) = -\frac{1}{z}$$

generate the $q = 4$ Hecke group. Section 3 verifies the relation $(JT)^4 = 1$ directly from the context matrices and formulates the resulting interface from arithmetic words to group operators and then to cell cosets. This restricted alphabet is important because its hyperbolic cellulations are produced by arithmetic operator semantics rather than imposed after endpoint evaluation. The interface is deliberately not a bijection: relations identify different histories, and stabilizers identify different operators at the level of cells or endpoints.

Projective continuation does not repair an inadmissible ordinary computation. For example, c/z has a well-defined projective value at $z = 0$, namely infinity, while ordinary division by zero remains undefined. We keep the ordinary domain and the projective extension as separate data throughout.

1.3 Why branching histories and order topology

The sequential fragment is the boundary case in which the dependency poset is a chain. A general branching tree has incomparable internal operations, hence many legal evaluation orders; these orders are exactly the maximal chains of a finite distributive lattice, the lattice of down-sets (partial evaluations). Domain theory then supplies the completion: the prefix order on words and the down-set order on computation DAGs each have an ideal completion into a dcpo whose compact elements are the finite histories. The Scott topology of this dcpo records "finitely observable" properties — those a finite prefix can certify — while the Lawson topology adjoins the dual side through the reciprocal involution, turning the one-sided order into a two-sided Hausdorff space. This is the order-topological carrier on which the analysis below is defined, and it is what makes "continuity of evaluation" a well-posed question rather than a slogan.

1.4 The analysis of histories

The histories carry a three-pillar analysis. The differential pillar turns the discrete generators into the exact flows of the two horizontal contact fields and identifies the order defect of the minimal branching DAG with the contact curvature $[D_u, D_v] = \mu\lambda\,\partial_a$. The integral pillar reads the same defect as a group-valued 1-cocycle on the linear-extension graph, closed by the Jacobi identity, and as the boundary holonomy of a holographic Stokes theorem. The series pillar expands the

history evaluation into the free Lie algebra as a Magnus series, whose arity-k brackets are carried by the $(k-1)$-cells of the down-set lattice and whose abelianization is the difference substitution of automated inequality proving. These three pillars are the order-topological reading of the geometric contact structure imported from Paper 0.

1.5 The hole–cap geometry

On the geometric side, the "holes" of the theory are of five distinct kinds and the "boundaries" of four; a decorated network thickened to a ball-and-tube body makes the distinction concrete. A surface hole is not filled inside the surface but is filled by an interior disk, and cutting along that disk exposes two caps. A constant-boundary disk admits no regular AES capping, which is why port substitution, topological filling, regular collar descent, and certified sealing must be kept apart.

1.6 Main results

The first group of results is combinatorial and syntactic. The trees with a unique legal internal evaluation order are exactly those whose internal vertices form a single spine; marking a leaf at the innermost vertex turns such a tree into a bounded marked spinal history, and the two constructions are mutually inverse when planar data are retained. Operand-slot chirality is part of that data and is kept distinct from chronological direction.

The second group is projective. The non-degenerate bilateral contexts generate $\mathrm{PGL}_2(K)$ for every field K, the contexts fixing infinity form the affine sector, and the case division in that generation proof yields the rank-one Bruhat decomposition. The generation theorem is Paper 0's, where it is proved for arbitrary fields; the statement and the Bruhat consequence are repeated here at the history level because the rest of the paper is written in that language. The $\mathbb{Q}(\sqrt{2})$-sublanguage generated by translation by $\sqrt{2}$ and by inversion carries the order-four relation $(JT)^4 = 1$, realized by an explicit eight-step marked history that is projectively trivial and ordinarily inadmissible at one initial value. That example is the reason this paper keeps ordinary admissibility as a separate hypothesis from projective non-degeneracy.

The third group is order-topological. For a branching tree the dependency poset carries a down-set information order whose ideal completion is a dcpo of finite and infinite computation DAGs, and the partial evaluations of a fixed tree form a finite distributive lattice whose maximal chains are the spinal histories. On the continuous lattice $[0, \infty]$ the three elementary contexts are the Scott–Lawson primitives: addition witnesses the way-below relation, scaling is a Scott automorphism, and the reciprocal is the order-reversing involution whose adjoining turns the one-sided Scott topology into the two-sided Hausdorff Lawson topology, mirroring the passage from the positive affine semigroup to $\mathrm{PGL}_2(\mathbb{R})$. The projective evaluation map has no Scott-continuous extension to the infinite histories: its only candidate limit is the infinite matrix product, i.e. the continued fraction, which pins down the finite-history scope of the formalism.

The fourth group is the analysis of histories, in three pillars. The differential pillar identifies the order defect of the minimal branching DAG $\{\mathrm{add}, \mathrm{multiply}\}$ with the contact curvature $[D_u, D_v] = \mu\lambda\,\partial_a$. The integral pillar expresses the order defect as a group-valued 1-cocycle on the linear-extension graph, closed by the Jacobi identity and by the holographic Stokes theorem. The series pillar expands the history evaluation into the free Lie algebra as a Magnus series, whose arity-k brackets are carried by the $(k-1)$-cells of the down-set lattice and whose abelianization is the difference substitution.

The fifth group is the hole–cap geometry. Five notions of hole and four of boundary are kept distinct; a decorated network thickened to a ball-and-tube body exhibits the hole–disk complementarity,

and a constant-boundary disk admits no regular AES capping.

The sixth group is the zero theory of regular arithmetic expression spaces. Section 7 restates the definition imported from Paper 0; Section 8 proves that the zero locus of a regular assignment is a smooth curve, that a complete connected boundaryless regular AES splits as $Z(a) \times \mathbb{R}$, that this forbids branching or multiple zero components in the regular category, and that a smooth parameter family with spatially regular zeros has a submersive total zero set — with the properness warning that prevents the last statement from being read as triviality. The section then defines singular arithmetic expression spaces and verifies one isolated-zero model in detail.

Everything else that earlier versions of this paper developed — the two affine translation coordinates, the continuous arithmetic motion, the basic hyperbolic model $\mathfrak{E}_0$, the accumulative commutative space, tearing, and the contact structure — remains developed in Paper 0, where it is proved at the same labels. Per the amendment in `governance/00d-paper-I-calculus-and-capping-amendment.md`, this paper now restates, at the same labels, the contact model, the ACS area formula, and the punctured-AES collar contract it uses, and develops the order-topological and analytic structure and the hole–cap geometry as its own chapters.

1.7 Scope and relation to standard geometry

Several ingredients used here are standard: finite posets and their linear extensions, fractional linear transformations, and the regular-value theorem. The mathematical contribution sought in Paper I is their history-sensitive arithmetic organization and the exact interfaces among them.

Four boundaries are enforced. First, the geometric construction of arithmetic expression spaces, their motions, their commutativization, and their contact structure belongs to Paper 0. Second, a horizontal contact distribution does not by itself select a unique complex structure, so arithmetic holomorphic and harmonic theories are deferred to Paper II; this includes any automorphic-function theory built on the G_4 action. Third, the regular-value theorem is only the starting point for singular families: pulling analytic real loci back to the Hecke cellulation, and developing their multiple zeros, discriminants, tubes, braids, and knots, belong to Paper III. Fourth, projective bivaluations, homogeneous quotient towers, and condensation belong to Paper IV, and no computational lower bound follows from noncommutativity or negative curvature alone.

1.8 Organization

Section 2 develops sequential and branching trees and marked histories, the down-set dcpo, and Birkhoff duality. Section 3 gives the bilateral projective semantics, the affine sector, and the Scott–Lawson topological correspondence. Section 4 develops the differential pillar, Section 5 the integral pillar, and Section 6 the series pillar. Section 7 restates the regular-AES interface imported from Paper 0. Section 8 develops the taxonomy of holes and the regular and singular zero geometry, and Section 9 the theory of caps and interface disks. Section 10 records what is proved, what remains open, and the interfaces to the later papers. The appendices collect convention checks, the equality-level examples, the Magnus-series symbolic verification, and the pants-cellulation homology verification.

2 Sequential Arithmetic Histories

Arithmetic evaluation is usually presented as a map from a syntactic expression to its value. This description suppresses the order in which intermediate values are formed. We begin by isolating the finite binary trees for which the internal operations have only one legal temporal order. Such a

tree is not yet a history: one must also mark which leaf of its innermost operation is to be carried forward as the evolving value. The resulting object, a marked spinal history, is the syntactic unit used in the remainder of the paper.

Throughout this section, K is a field and

$$\Omega = \{+, -, \times, /\}$$

is the binary arithmetic signature. Division is interpreted as a partial operation in ordinary arithmetic. The combinatorial statements below apply to any binary signature; the choice of Ω is needed only when histories are evaluated.

2.1 Expression trees and dependency

Definition 2.1 (Arithmetic expression tree)**.** An *arithmetic expression tree over* K is a finite rooted planar binary tree whose leaves are labelled by elements of K or by formal variables and whose internal vertices are labelled by elements of Ω. Its set of internal vertices is denoted by $I(T)$.

The planar structure records the first and second operand slots. It does not, by itself, specify a temporal order between operations on disjoint subtrees. That order is recorded by the dependency relation.

Definition 2.2 (Internal dependency poset)**.** For $u, v \in I(T)$, write $u \preccurlyeq_T v$ when either $u = v$ or the value produced at u is required as an input at v. Equivalently, v lies on the path from u to the root. A *legal internal evaluation order* is a linear extension of the finite poset $(I(T), \preccurlyeq_T)$.

Leaves are not included in $I(T)$. Thus a tree with no internal vertex has the unique empty internal evaluation order. This convention separates the order of operations from the prior availability of atomic inputs.

We shall use the following elementary finite-poset fact in both directions.

Lemma 2.3 (Unique linear extensions)**.** *A finite poset has a unique linear extension if and only if it is a chain.*

Proof. Every linear extension of a chain is its given total order, so a chain has a unique linear extension. Conversely, let P be a finite poset and suppose that $u, v \in P$ are incomparable. Adding the relation $u < v$ and taking its transitive closure cannot create a directed cycle: such a cycle would contain a pre-existing path $v \leq u$, contrary to incomparability. The resulting finite partial order has a linear extension. The same argument with $v < u$ gives a second linear extension. One orders u before v and the other orders v before u, so the extension of P is not unique. Hence a poset with a unique linear extension contains no incomparable pair and is a chain. □

Theorem 2.4 (Sequential-tree classification)**.** *For a finite binary expression tree T, the following conditions are equivalent.*

1. *The dependency poset $I(T)$ has a unique linear extension.*
2. *The dependency poset $I(T)$ is a chain.*
3. *Every internal vertex has at most one internal child.*
4. *The internal vertices form a single spine: either $I(T)$ is empty, or there is a path from the root to an innermost internal vertex containing every internal vertex of T.*

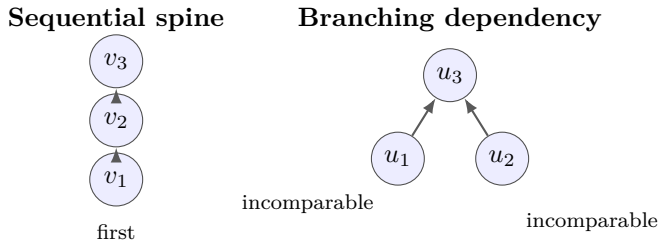


Figure 2: A sequential tree has a chain of internal dependencies (left). A branching tree contains incomparable internal operations and therefore admits more than one legal internal evaluation order (right).

Proof. The equivalence of the first two conditions is Lemma 2.3.

Suppose that $I(T)$ is a chain. If an internal vertex had two internal children, neither child could depend on the other, since they lie in disjoint subtrees. They would therefore be incomparable in $I(T)$, a contradiction. Thus (2) implies (3).

Assume (3). If $I(T)$ is empty there is nothing to prove. Otherwise begin at the root. Whenever the current internal vertex has an internal child, that child is unique. Following these children produces a path ending at an internal vertex with two leaf children. Every internal vertex lies on this path: an internal vertex off the path would occur in the other child-subtree of some vertex on the path, giving that vertex two internal children. Hence the path contains all internal vertices, proving (4).

Finally, if all internal vertices lie on one root-to-leaf path, their ancestor relation is total. Read from the innermost vertex towards the root, each operation depends on its predecessor. Thus $I(T)$ is a chain, proving (4)⇒(2) and completing the equivalence. □

Figure 2 contrasts the two possible local patterns at the level of internal dependencies. It is only a visual summary of the order relation; the classification rests on the preceding proof.

Definition 2.5 (Sequential tree)**.** A finite binary expression tree satisfying the equivalent conditions of Theorem 2.4 is called a *sequential tree*.

The adjective refers only to internal evaluation. It does not choose an active leaf, and it does not assert that ordinary arithmetic evaluation is defined. For example,

$$(a+b)(c+d)$$

is not sequential: the two additions are incomparable dependencies of the final multiplication. By contrast,

$$\big((a+b)c-d\big)/e$$

is sequential, because its internal vertices lie on one spine.

2.2 One-hole contexts and the marked accumulator

Let z denote a hole rather than an arithmetic variable. For $\omega \in \Omega$ and $c \in K$, define the two elementary one-hole contexts

$$C^{(1)}_{\omega,c}[z] = \omega(z,c), \qquad C^{(2)}_{\omega,c}[z] = \omega(c,z). \tag{2}$$

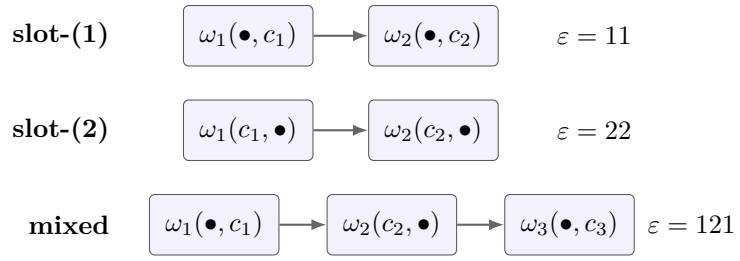


Figure 3: Pure slot-(1), pure slot-(2), and mixed marked histories. In each box, $\bullet$ is the current accumulator—the marked seed in the first box and the preceding output thereafter. Its displayed position is the operand slot, while time always runs along the arrows.

The superscript gives the operand slot occupied by the evolving value. It is therefore intrinsic to the marked process and should not be replaced by a description based on how a planar drawing happens to expand.

At the innermost internal vertex of a nontrivial sequential tree, both children are leaves. The dependency poset determines that this operation is performed first, but it does not determine which of its two inputs is the value to be carried along the spine. Choosing one of those leaves as the initial accumulator removes this ambiguity.

Definition 2.6 (Marked spinal history)**.** A *free marked spinal history over K* is a finite chronological word

$$\gamma = (g_1, \dots, g_n), \qquad g_i = C_{\omega_i, c_i}^{(\varepsilon_i)}, \quad \varepsilon_i \in \{1, 2\}.$$

The context g_1 is applied first and g_n last. Its *chirality word* is

$$\varepsilon(\gamma) = \varepsilon_1 \cdots \varepsilon_n.$$

A *bounded marked spinal history* is a pair $(x_0; \gamma)$ consisting of an initial accumulator x_0 and a free history. The empty word is allowed; its bounded form represents the marked single-leaf tree.

When no confusion is possible, we call either form a marked spinal history and state explicitly whether an initial value is present. Earlier drafts called certain pure-chirality cases threadlike expressions. The present definition also includes mixed chirality and makes the marking part of the data.

Figure 3 displays operand-slot chirality directly in the one-hole contexts. The diagram is a notation guide, not a substitute for the marked-tree correspondence proved below.

Proposition 2.7 (Marked-tree–history correspondence)**.** *Let T be a sequential tree all of whose leaves are labelled by elements of K.*

1. *If T is a single leaf, marking that leaf determines the empty bounded history.*
2. *If T is nontrivial, a choice of one leaf at its innermost internal vertex as the initial accumulator determines a unique bounded marked spinal history.*
3. *Conversely, a bounded marked spinal history determines a unique marked planar sequential tree.*

These two constructions are mutually inverse when operation labels, constants, operand slots, and the initial mark are retained.

Proof. The single-leaf case is immediate. Suppose T is nontrivial. By Theorem 2.4, its internal vertices form a unique spine. At the innermost vertex, mark one leaf x_0 and denote the other by c_1. The operation label and the slot containing x_0 determine $g_1 = C^{(\varepsilon_1)}_{\omega_1,c_1}$. Move one vertex towards the root. Its off-spine child must be a leaf, since otherwise it would be a second internal child. Its label c_2, the operation label, and the slot occupied by the spine determine g_2. Repeating this procedure reaches the root and produces a unique chronological word $(g_1, \dots, g_n)$.

Conversely, start with a marked leaf x_0. Replace it by the planar binary tree $C^{(\varepsilon_1)}_{\omega_1,c_1}[x_0]$, then place the resulting tree in slot ε_2 of the next context, and continue. Exactly one child at every new internal vertex is internal, so the result is sequential. Each step records the operation, off-spine leaf, and slot used in the forward construction. The constructions therefore recover one another. □

The proposition concerns fully labelled planar trees. If parentheses, constant labels, or the marked seed are forgotten, the correspondence is no longer injective.

2.3 Composition and ordinary evaluation

If

$$\gamma = (g_1, \dots, g_m), \qquad \delta = (h_1, \dots, h_n),$$

their chronological concatenation is

$$\gamma\delta = (g_1, \dots, g_m, h_1, \dots, h_n), \tag{3}$$

so γ is performed first and δ second. This convention is fixed for the entire paper and is summarized again in Appendix A.

Each elementary context determines a partial function $g_i \colon K \dashrightarrow K$. Starting from $x \in K$, define recursively

$$x_0 = x, \qquad x_i = g_i(x_{i-1}).$$

The bounded history is *ordinarily admissible at* x when every x_i is defined in ordinary field arithmetic. In that case its endpoint is

$$\nu_x(\gamma) = g_n \circ \cdots \circ g_1(x). \tag{4}$$

The admissible domain of a free history is the set of such x. In particular, a second-slot division c/z excludes any state at which the evolving value is zero, while a first-slot division z/c requires $c \neq 0$.

The concatenation convention gives

$$\nu_x(\gamma\delta) = \nu_{\nu_x(\gamma)}(\delta) \tag{5}$$

whenever both sides are ordinarily admissible. Equation (5) is a statement about partial functions. Chapter 3 introduces a projective evaluation with a different domain convention.

2.4 Mirror, temporal reversal, and path inverse

Three operations on a context word must be kept separate.

Definition 2.8 (Mirror and temporal reversal)**.** For $g = C^{(\varepsilon)}_{\omega,c}$, let

$$m(g) = C^{(3-\varepsilon)}_{\omega,c}.$$

The *mirror* and *temporal reversal* of $\gamma = (g_1, \dots, g_n)$ are respectively

$$m\gamma = (m(g_1), \dots, m(g_n)), \qquad r\gamma = (g_n, \dots, g_1).$$

The mirror exchanges operand slots while retaining chronological order; temporal reversal reverses chronological order while retaining the contexts.

Both operations are involutions on marked words, and they commute: $m(r\gamma) = r(m\gamma)$. This is a syntactic statement; ordinary domains of the evaluated words need not agree.

Definition 2.9 (Path inverse)**.** Suppose every context g_i in γ acts as an invertible function on a declared domain. The *path inverse* is the functional word

$$\gamma^{-1} = (g_n^{-1}, \ldots, g_1^{-1}).$$

It is defined only under this invertibility hypothesis and need not have the same operation labels or ordinary admissible domain as $r\gamma$.

For example, let $g_1[z] = z + 1$ and $g_2[z] = 2z$. Then

$$\gamma = (g_1, g_2)\colon x \longmapsto 2x + 2.$$

Because addition and multiplication are symmetric operations, $m\gamma$ has the displayed expression $2(1 + x)$ and induces the same operator $2x + 2$. In contrast,

$$r\gamma = (g_2, g_1)\colon x \longmapsto 2x + 1,$$

while

$$\gamma^{-1} = (z \mapsto z/2,\ z \mapsto z - 1)\colon x \longmapsto x/2 - 1.$$

Thus equality with a mirror in this example is caused by the symmetry of the chosen elementary operations; it does not identify mirror, reversal, and inverse. The affine order defect studied later compares temporal orders. It is not automatically an invariant of planar mirror.

2.5 Information retained and information forgotten

Several equivalence relations occur naturally, and the remainder of the paper uses them at different stages. It is useful to order them by the information they forget.

1. *Literal syntactic equality* retains the fully parenthesized written expression.
2. *Marked-history equality* retains the initial mark, the chronological context word, and every operand-slot label. It may forget inessential typography but not process data.
3. *Operator equality* identifies histories inducing the same partial function on a stated ordinary domain, or the same projective transformation after projective evaluation.
4. *Charge equality*, when a charge map has been specified, retains only the chosen abelian totals of a history.
5. *Endpoint equality* at x asks only whether the admissible values $\nu_x(\gamma)$ and $\nu_x(\delta)$ coincide.
6. *Quotient or relational equality* is any further identification imposed by an explicitly declared rewriting or condensation relation.

This list is not a claim that every adjacent pair gives a strict quotient in all languages. It is a discipline for stating which level is being used. For example, the two histories

$$\gamma = (z \mapsto z + 1,\ z \mapsto 2z), \qquad \delta = (z \mapsto 3z,\ z \mapsto z + 1)$$

are distinct marked histories and induce the distinct operators $2z + 2$ and $3z + 1$, but at the initial value $x = 1$ both have endpoint 4.

2.6 Representative chirality patterns

The slot notation exposes examples that the older comb terminology obscured.

Example 2.10 (Pure slot-(1) history)**.** The word

$$C^{(1)}_{+,2}, \quad C^{(1)}_{\times,3}, \quad C^{(1)}_{-,4}$$

has chirality word 111 and evaluates ordinarily as $x \mapsto (x+2)3-4$.

Example 2.11 (Pure slot-(2) history)**.** The word

$$C^{(2)}_{-,2}, \quad C^{(2)}_{/,3}$$

has chirality word 22 and evaluates as

$$x \longmapsto \frac{3}{2-x},$$

on the ordinary domain $K \setminus \{2\}$. The accumulator occupies the second operand slot at both steps.

Example 2.12 (Mixed chirality)**.** The word

$$C^{(1)}_{-,1}, \quad C^{(2)}_{/,2}, \quad C^{(1)}_{+,3}$$

has chirality word 121 and evaluates as

$$x \longmapsto \frac{2}{x-1} + 3,$$

on its ordinary domain. Its dependency poset is still a chain; mixed chirality does not introduce branching.

The syntactic passage is therefore

$$\begin{gathered}\text{unmarked sequential tree} \xrightarrow{\text{choice of accumulator}} \text{marked planar sequential tree},\\ \text{marked planar sequential tree} \longleftrightarrow \text{bounded marked spinal history},\\ \text{bounded marked spinal history} \longrightarrow \text{operator} \longrightarrow \text{endpoint}.\end{gathered}$$

when the history is bounded and all planar labels are retained. The first arrow is a choice and the double arrow is a bijection; only the later evaluation arrows forget process data. Chapter 3 constructs the operator stage for all non-degenerate bilateral contexts.

2.7 The prefix information order and ideal completion

The chronological concatenation of (3) makes the free marked spinal histories Γ^* a monoid. Domain theory turns this monoid into an ordered and then a complete object. Throughout, $\Gamma = \{\, C^{(\varepsilon)}_{\omega,c} \,\}$ denotes the alphabet of elementary one-hole contexts.

Definition 2.13 (Prefix information order)**.** For free marked spinal histories $\gamma, \gamma' \in \Gamma^*$, write $\gamma \sqsubseteq \gamma'$ when $\gamma' = \gamma\delta$ for some $\delta \in \Gamma^*$. This is the *chronological prefix order*: γ is an earlier truncation of γ', retaining strictly less process data.

The prefix order is a partial order, but Γ^* is not directed complete: an infinite chain $\gamma_0 \sqsubseteq \gamma_1 \sqsubseteq \cdots$ has no join in Γ^*. Its completion is the classical ideal completion of a poset.

Proposition 2.14 (Spinal ideal completion)**.** *The ideal completion of* $(\Gamma^*, \sqsubseteq)$ *is*

$$\mathrm{Idl}(\Gamma^*) \;\cong\; \Gamma^{\leq\omega} := \Gamma^* \cup \Gamma^\omega,$$

the dcpo of finite and countably infinite chronological words, ordered by prefix. Its compact elements are exactly the finite words Γ^**, and each infinite word* $\gamma_\infty = (g_1, g_2, \ldots)$ *is the directed join of its finite prefixes:*

$$\gamma_\infty = \bigsqcup_{n\geq 0} (g_1, \ldots, g_n).$$

Proof. The ideals of a prefix-ordered free monoid are exactly the principal ideals $\downarrow\gamma$ for finite γ and the directed families of all finite prefixes of an infinite word; the bijection sends an infinite word to the directed union of its prefixes. This is the standard ideal-completion construction for the domain of finite and infinite words [6, 2]. □

2.8 Branching histories and the down-set information order

The prefix order is the special case in which the dependency poset is a chain. For a general (branching) tree the internal dependency poset $P = (I(T), \preccurlyeq_T)$ is not a chain, and the natural information order is the *down-set* (order-ideal) embedding.

Definition 2.15 (Down-set information order)**.** Let FinDAG be the class of finite computation DAGs: finite dependency posets P whose vertices carry operation, constant, and operand-slot decorations, as in Definition 2.2. For $P, Q \in \mathrm{FinDAG}$, write $P \sqsubseteq Q$ when P is a decorated *down-set* (order ideal) of Q: $P \subseteq Q$, the order of P is the restriction of that of Q, and

$$x \in P,\; y \in Q,\; y \leq_Q x \implies y \in P.$$

Read "$P \sqsubseteq Q$" as "P is a completed sub-computation of Q".

Directed families under $\sqsubseteq$ are towers of sub-computations, and their direct union is a well-defined (possibly infinite) poset. The completion is again classical.

Theorem 2.16 (Branching DAG dcpo)**.** *The ideal completion* $\mathrm{Idl}(\mathrm{FinDAG}, \sqsubseteq)$ *is isomorphic to the class of computation DAGs in which every vertex has only finitely many predecessors (equivalently, DAGs that are the directed union of their finite down-sets). The compact elements are exactly the finite computation DAGs, and each such infinite DAG* Q *is the directed join of its finite down-sets:*

$$Q = \bigsqcup \{\, P \subseteq Q : P \text{ finite down-set} \,\}.$$

A "broom" DAG — infinitely many minimal vertices feeding a single top vertex — is excluded, because its top vertex has infinitely many predecessors and lies in no finite down-set. When P *is a chain (a sequential tree), the down-set order restricts to the prefix order of Definition 2.13.*

Proof. An ideal I is a directed, downward-closed family of finite DAGs; its direct union $\bigcup I$ is well-defined, and every element of it lies in some finite member of I (hence has finitely many predecessors). Conversely, a DAG in which every vertex has finitely many predecessors is the direct union of its finite down-sets, and that family is an ideal. The two maps are inverse. The excluded-broom observation is immediate, and the chain specialization is the bijection between the down-sets of a chain and its prefixes. □

2.9 Partial evaluations and Birkhoff duality

For a fixed finite dependency poset P, the partial evaluations — the completed sub-computations — form a finite distributive lattice, and the legal evaluation orders are its maximal chains.

Theorem 2.17 (Down-set lattice and spinal histories)**.** *Let $P = I(T)$ be the internal dependency poset of a finite expression tree T, and let $\mathcal{O}(P)$ be its set of down-sets ordered by inclusion.*

1. *$\mathcal{O}(P)$ is a finite distributive lattice — the lattice of partial evaluations of T — and by Birkhoff duality* $\mathrm{Stone}(\mathcal{O}(P)) \cong P$ *(the join-irreducibles recover the dependency poset).*

2. *The legal internal evaluation orders of T are in bijection with the maximal chains*

$$\varnothing = D_0 \subsetneq D_1 \subsetneq \cdots \subsetneq D_{|P|} = P$$

   of $\mathcal{O}(P)$, each step adjoining one minimal vertex.

3. *After a choice of initial accumulator and operand-slot marking, each maximal chain encodes exactly one marked spinal history. A sequential tree is the case in which $\mathcal{O}(P)$ is a single chain; a branching tree has $\mathcal{O}(P)$ a nontrivial distributive lattice, and its complete evaluation space is organized by $\mathcal{O}(P)$ together with its maximal chains.*

Proof. Part (1) is Birkhoff's representation theorem. For part (2), a maximal chain from $\varnothing$ to P must add exactly one vertex at a time, and the added vertex is minimal in the complement, so the sequence of additions is precisely a linear extension of P; the converse is direct. Part (3) reads that linear extension through the marked-tree–history correspondence of Proposition 2.7. □

3 Projective Semantics and the Affine Sector

The four arithmetic operations behave asymmetrically once the accumulator may occupy either operand slot. Most elementary contexts are affine, but the second-slot division c/z is genuinely projective. This section makes that enlargement precise. The projective semantics does not replace ordinary arithmetic evaluation: it supplies an operator-valued continuation on a different domain.

Throughout this section, K is an arbitrary field. Points of the projective line are one-dimensional subspaces of K^2, written in homogeneous coordinates as

$$[u : v], \qquad (u, v) \neq (0, 0).$$

We use the affine chart $z = [z : 1]$ and write $\infty = [1 : 0]$. Matrices act on column vectors from the left. Thus

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \quad \Longrightarrow \quad M \cdot z = \frac{Az + B}{Cz + D}, \tag{6}$$

with the homogeneous interpretation at poles and at infinity. Replacing M by a nonzero scalar multiple does not change the projective map.

3.1 Ordinary domains and projective continuation

For an elementary context $g = C^{(\varepsilon)}_{\omega,c}$, ordinary evaluation gives a partial function $g\colon K \dashrightarrow K$. A *non-degenerate projective context* is one for which the corresponding homogeneous matrix is invertible. It acts everywhere on $\mathbb{P}^1(K)$, although its affine-coordinate formula may have a pole.

The elementary contexts and matrix representatives are as follows.

Context	Affine-coordinate map	Matrix representative
$C^{(1)}_{+,c} = C^{(2)}_{+,c}$	$z \mapsto z + c$	$\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$
$C^{(1)}_{-,c}$	$z \mapsto z - c$	$\begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix}$
$C^{(2)}_{-,c}$	$z \mapsto c - z$	$\begin{pmatrix} -1 & c \\ 0 & 1 \end{pmatrix}$
$C^{(1)}_{\times,c} = C^{(2)}_{\times,c}$	$z \mapsto cz$	$\begin{pmatrix} c & 0 \\ 0 & 1 \end{pmatrix}$
$C^{(1)}_{/,c}$	$z \mapsto z/c$	$\begin{pmatrix} 1 & 0 \\ 0 & c \end{pmatrix}$
$C^{(2)}_{/,c}$	$z \mapsto c/z$	$\begin{pmatrix} 0 & c \\ 1 & 0 \end{pmatrix}$

The first three rows are invertible for every $c \in K$. Multiplication by c, first-slot division by c, and second-slot division with numerator c are projectively invertible precisely when $c \neq 0$. At $c = 0$, multiplication and $0/z$ are constant on their ordinary domains, while $z/0$ has empty ordinary domain. Their singular matrix representatives do not act everywhere on $\mathbb{P}^1(K)$: each has a projective base point. They are therefore excluded from $\mathrm{PGL}_2(K)$ and from the non-degenerate history language used below.

The table also displays the distinction between ordinary and projective domains. If $c \neq 0$, ordinary first-slot division z/c is defined on all of K. Ordinary second-slot division c/z is defined only for $z \neq 0$. Projectively, the latter sends 0 to ∞ and ∞ to 0. This is a valid action on $\mathbb{P}^1(K)$ but not an ordinary evaluation through a division by zero.

Warning 3.1 (Projective values are not ordinary values). Projective continuation records a fractional linear operator and its chart transitions. It does not certify that the original arithmetic word is ordinarily admissible at a given initial value. Whenever an endpoint in K is used as an ordinary arithmetic result, ordinary admissibility must be checked separately.

3.2 Projective evaluation of a chronological history

Let $\mathrm{Hist}^{\mathrm{nd}}_K$ denote the free marked spinal history language generated by the non-degenerate elementary contexts in the table. The superscript "nd" refers to projective non-degeneracy, not to ordinary admissibility at every affine input.

Definition 3.2 (Projective evaluation)**.** Choose for each elementary non-degenerate context g_i a matrix representative $M_i \in \mathrm{GL}_2(K)$. For the chronological word $\gamma = (g_1, \ldots, g_n)$, with g_1 applied first, define

$$\rho(\gamma) = [M_n \cdots M_1] \in \mathrm{PGL}_2(K). \tag{7}$$

For the empty word, set $\rho(\varnothing) = 1$.

This definition uses projective operator evaluation. For a bounded history $(x; \gamma)$, its projective endpoint is $\rho(\gamma) \cdot x$ in $\mathbb{P}^1(K)$. Its ordinary endpoint $\nu_x(\gamma)$ remains the partial evaluation defined in Section 2.

Proposition 3.3 (Compatibility with chronology)**.** *Projective evaluation is independent of the chosen scalar matrix representatives. If $\gamma\delta$ means first γ and then δ, then*

$$\rho(\gamma\delta) = \rho(\delta)\rho(\gamma). \tag{8}$$

Moreover, whenever γ is ordinarily admissible at x and its ordinary endpoint lies in K, the projective endpoint agrees with it in the affine chart:

$$\rho(\gamma) \cdot x = [\nu_x(\gamma) : 1].$$

Proof. Replacing M_i by $a_i M_i$, with $a_i \in K^\times$, multiplies the product $M_n \cdots M_1$ by the nonzero scalar $\prod_i a_i$ and hence does not alter its projective class. If $\gamma = (g_1, \ldots, g_m)$ and $\delta = (h_1, \ldots, h_n)$, then the matrix word for $\gamma\delta$ is

$$N_n \cdots N_1 M_m \cdots M_1,$$

which represents $\rho(\delta)\rho(\gamma)$. This proves (8).

For the last assertion, each row of the table agrees with its ordinary arithmetic context wherever that context is defined. Applying this observation inductively along the admissible intermediate states proves equality in the affine chart. □

Thus ρ is a homomorphism from chronological words if the target group is written with the opposite multiplication. We retain ordinary matrix multiplication and the explicit rule (8), because it makes both the column action and function composition standard.

3.3 Generation of the projective group

Introduce the elementary projective transformations

$$T_s(z) = z + s, \qquad D_q(z) = qz \quad (q \in K^\times), \qquad J(z) = -\frac{1}{z}.$$

They are projective evaluations of, respectively, addition by s, multiplication by q, and second-slot division with numerator -1. These three kinds of context suffice for the entire bilateral projective language at the operator level.

Theorem 3.4 (Bilateral arithmetic generates PGL_2)**.** *Let K be any field. The projective evaluations of non-degenerate bilateral four-operation marked spinal histories have image*

$$\rho(\mathrm{Hist}_K^{\mathrm{nd}}) = \mathrm{PGL}_2(K).$$

Equivalently, translations T_s, nonzero scalings D_q, and J generate $\mathrm{PGL}_2(K)$.

Proof. The inclusion $\rho(\mathrm{Hist}_K^{\mathrm{nd}}) \subseteq \mathrm{PGL}_2(K)$ holds by definition: every elementary context admitted to $\mathrm{Hist}_K^{\mathrm{nd}}$ has an invertible matrix representative, and a product of invertible matrices is invertible.

For the converse, each of T_s, D_q, and J is an elementary non-degenerate context of the table above, so the image contains the subgroup they generate. That subgroup is all of $\mathrm{PGL}_2(K)$ by the generation theorem for projective arithmetic, which is Theorem 6.2 of Paper 0 and is proved there for an arbitrary field without any ordering or characteristic hypothesis. Reading that theorem through the present history language gives the stated equality. □

The generation statement is therefore a theorem of the series rather than of this paper alone: its proof, and the constructive factorization it yields, are in Paper 0. What belongs to this paper is the reading of that factorization as a chronological word. Paper 0's proof is constructive, giving a history of length at most four for any specified projective matrix once arbitrary field constants are admitted as context parameters. It establishes surjectivity at the operator level, not uniqueness of a representing history. In fact, many distinct context words have the same projective evaluation.

Example 3.5 (A pole in an ordinarily inadmissible history)**.** Let $K = \mathbb{Q}$ and consider the one-step history $g[z] = 1/z$. It is a non-degenerate projective context with $\rho(g) \cdot 0 = \infty$. The bounded history $(0; g)$ has no ordinary endpoint. Appending another inversion gives a projective word acting as the identity, but the two-step ordinary evaluation at 0 is still inadmissible because its first step is undefined.

3.4 A distinguished arithmetic sublanguage: the q=4 Hecke group

The surjectivity theorem allows arbitrary constants and produces the whole projective group. A much smaller constant alphabet already carries a useful discrete geometry. Work in $\mathbb{Q}(\sqrt{2}) \subset \mathbb{R}$, and let Hist_4 be the symmetric marked history language generated by the three elementary contexts

$$\mathsf{t}_+[z] = z + \sqrt{2}, \qquad \mathsf{t}_-[z] = z - \sqrt{2}, \qquad \mathsf{j}[z] = -\frac{1}{z}.$$

Here "symmetric" means that the inverse translation is retained as a generator; j is its own projective inverse. Write

$$T = T_{\sqrt{2}}, \qquad G_4 = \rho(\mathrm{Hist}_4) = \langle T, J \rangle \leq \mathrm{PGL}_2(\mathbb{R}).$$

This is the standard projective realization of the $q = 4$ Hecke group [5, 3].

Proposition 3.6 (The order-four arithmetic relation)**.** *With the column action and chronological convention fixed above, the generators have determinant-one representatives*

$$\widetilde{T} = \begin{pmatrix} 1 & \sqrt{2} \\ 0 & 1 \end{pmatrix}, \qquad \widetilde{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \textit{in } \mathrm{SL}_2(\mathbb{Z}[\sqrt{2}]).$$

Consequently G_4 is defined by matrices over $\mathbb{Z}[\sqrt{2}]$. In $\mathrm{PGL}_2(\mathbb{R})$ one has

$$J^2 = 1, \qquad (JT)^4 = 1, \tag{9}$$

and JT has exact projective order four. The second relation is the projective evaluation of the chronological eight-step history

$$\omega_4 = (\mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}, \mathsf{t}_+, \mathsf{j}). \tag{10}$$

Thus ω_4 and the empty history are distinct marked histories with the same projective operator.

Proof. Both displayed matrices have determinant one and entries in $\mathbb{Z}[\sqrt{2}]$; their inverses have the same property. Moreover,

$$\widetilde{J}^2 = -I, \qquad R := \widetilde{J}\widetilde{T} = \begin{pmatrix} 0 & -1 \\ 1 & \sqrt{2} \end{pmatrix}, \qquad R^2 = \begin{pmatrix} -1 & -\sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}.$$

A second multiplication gives $R^4 = -I$. Hence the scalar classes satisfy (9). Neither R nor R^2 is scalar, so the projective order of JT is exactly four. Finally, each chronological pair $(\mathsf{t}_+, \mathsf{j})$ evaluates as $J \circ T$ and is represented by $\widetilde{J}\widetilde{T} = R$. Four such pairs therefore evaluate as R^4, which is the identity in the projective group. □

The proposition is an operator statement, not a universal statement about ordinary real arithmetic. Every letter of Hist_4 is projectively non-degenerate, but a bounded history is ordinarily admissible only if the evolving affine value is nonzero whenever j is applied. For example, ω_4 is projectively neutral but is not ordinarily admissible at $x = -\sqrt{2}$: its first translation reaches zero, after which j requests division by zero. In contrast, the same matrices act without an interior pole on the upper half-plane $\mathbb{H}^2$. The Fuchsian action and ordinary evaluation on a specified real input are therefore two related but distinct uses of the same context word.

Remark 3.7 (From histories to Hecke cells). The conventional action of G_4 on $\mathbb{H}^2$ supplies its Hecke tessellations and derived dual cellulations. Paper I needs only the following formal interface. If $\mathcal{C}$ is a chosen cell in such a G_4-equivariant cellulation and $H_{\mathcal{C}} = \mathrm{Stab}_{G_4}(\mathcal{C})$, then the cell reached by a history factors as

$$\mathrm{Hist}_4 \xrightarrow{\rho} G_4 \longrightarrow G_4/H_{\mathcal{C}}, \qquad \gamma \longmapsto \rho(\gamma)H_{\mathcal{C}}. \tag{11}$$

Neither arrow is generally injective. Relations such as (10) identify distinct histories at the operator level, and distinct operators in one coset act on $\mathcal{C}$ as the same cell. Passing instead to one endpoint uses the stabilizer of that point and forgets still more information. Thus arithmetic histories, projective operators, cells, and endpoints must not be placed in one-to-one correspondence. The order-four relation provides an arithmetic entry point for the fourfold incidence visible in the $q = 4$ geometry; the construction of analytic functions and singular zero sets on that cellulation is deferred to Papers II and III.

3.5 The affine syntactic sector

Let B_∞ be the stabilizer of the chosen point at infinity:

$$B_\infty = \mathrm{Stab}_{\mathrm{PGL}_2(K)}(\infty).$$

An element represented by $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ fixes $\infty = [1:0]$ exactly when $C = 0$. After rescaling by D^{-1}, it then has a unique representative

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}, \qquad a \in K^\times, \quad b \in K,$$

acting as $z \mapsto az + b$.

Define $\mathrm{Hist}_K^{\mathrm{aff}}$ to be the syntactic sublanguage generated by those *elementary non-degenerate contexts* in the table that individually fix ∞. Equivalently, it excludes the elementary second-slot divisions c/z and retains every non-degenerate elementary affine context. This definition is deliberately syntactic: it does not assert that $\mathrm{Hist}_K^{\mathrm{aff}}$ is the entire inverse image $\rho^{-1}(B_\infty)$. A word containing projective steps can leave the affine chart and later evaluate to an affine operator.

Corollary 3.8 (Affine/Borel sector)**.** *For every field K,*

$$\rho(\mathrm{Hist}_K^{\mathrm{aff}}) = B_\infty \cong \mathrm{Aff}(1, K).$$

In particular, the affine syntactic sublanguage maps surjectively onto the Borel subgroup stabilizing ∞.

Proof. Every generator of $\mathrm{Hist}_K^{\mathrm{aff}}$ fixes ∞; therefore every product of their projective matrices fixes ∞, and the image is contained in B_∞. Conversely, every element of B_∞ acts as $z \mapsto az + b$ with $a \neq 0$. It is

$$T_b \circ D_a,$$

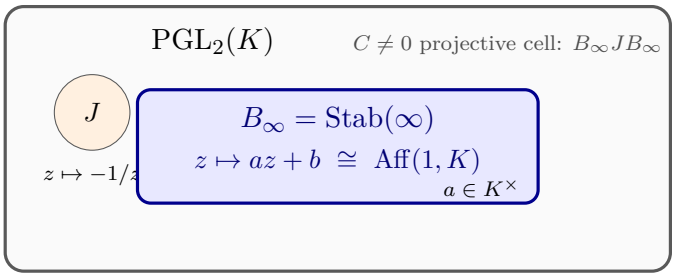


Figure 4: The affine/Borel sector inside the bilateral projective operator semantics. Elementary affine contexts generate the stabilizer B_∞, whereas inversion reaches the complementary Bruhat cell. Words containing projective steps may nevertheless evaluate back into B_∞.

the evaluation of a chronological word consisting of the non-degenerate affine contexts D_a followed by T_b. Hence the image is all of B_∞. The displayed normal form also identifies its group law with that of $\mathrm{Aff}(1, K)$. □

Figure 4 records the resulting operator-level placement. It does not identify the affine syntactic language with the full inverse image of the inner region.

The corollary locates the existing add–multiply language without collapsing the larger syntax. Pure slot-(1) non-degenerate histories lie in this affine sector, but so do second-slot subtraction and multiplication contexts. The criterion is stabilization of ∞, not the slot bit by itself.

3.6 The positive real component

Later differential-geometric sections use real multiplication in the form

$$z \longmapsto e^M z, \qquad M \in \mathbb{R}.$$

Its multiplier is positive. Together with all real translations, these maps form

$$\mathrm{Aff}^+(1, \mathbb{R}) = \{z \mapsto az + b : a > 0\},$$

the identity component of the real affine group. They do not include affine reflections with negative multiplier. The algebraic statement of Corollary 3.8 concerns all nonzero multipliers in $K^\times$; the continuous theory is its positive real component.

3.7 Bruhat placement and the projective infinitesimal direction

The case division in the proof of Paper 0, Theorem 6.2 also gives the rank-one Bruhat decomposition

$$\mathrm{PGL}_2(K) = B_\infty \sqcup B_\infty J B_\infty,$$

which Paper 0 records in the same chapter. The first cell is the case $C = 0$; the explicit factorization obtained for $C \neq 0$ lies in the second. The same dichotomy separates the contexts of the table above: the affine rows fix ∞, and the second-slot division c/z is the elementary step that leaves the first cell. This observation places bilateral arithmetic as a projective completion of the affine sector; no representation-theoretic conclusions beyond this decomposition are needed here.

There is also an infinitesimal counterpart over $\mathbb{R}$ (or $\mathbb{C}$). The projective vector fields

$$\partial_z, \qquad z\partial_z, \qquad z^2\partial_z$$

span the local projective Lie algebra. The first two generate affine flow. A conjugated translation detects the third direction:

$$J \circ T_s \circ J^{-1}(z) = \frac{z}{1 - sz} = z + sz^2 + O(s^2).$$

Consequently a general scalar projective flow has Riccati form

$$\dot{z} = \beta + \alpha z + \kappa z^2.$$

Paper I develops only the affine slice $\kappa = 0$. No projective metric, projective arithmetic expression space, or projective contact theory is asserted here.

3.8 Scott–Lawson roles of the elementary contexts

The three elementary contexts of the table have a precise order-topological meaning on the continuous lattice $L = [0, \infty]$ with its usual order; this reading is the bridge from the algebraic side of this section to the analytic chapters that follow. Write σ, ω, and λ for the Scott, lower, and Lawson topologies of L, and identify L with the closure of $\mathbb{R}_{>0}$ in $\mathbb{P}^1(\mathbb{R})$.

Proposition 3.9 (Domain-theoretic roles of the generators)**.** *On $L = [0, \infty]$:*

1. *the additive translation $T_s = C^{(1)}_{+,s}$ $(s \geq 0)$ is a Scott-continuous embedding, not onto for $s > 0$, and is the way-below witness: for $a > 0$,*

$$a \ll b \iff a < b \iff \exists s > 0 : \ T_s(a) = a + s \leq b,$$

   while $0 \ll 0$; the forward orbits generate the Scott basis $\{(a, \infty] : a \in L\}$;

2. *the positive scaling $D_k = C^{(1)}_{\times,k}$ $(k > 0)$ is an order automorphism of $(L, \leq)$, hence a Scott homeomorphism;*

3. *the second-slot division (reciprocal) $\neg = C^{(2)}_{/,1} : a \mapsto 1/a$ (with $0 \leftrightarrow \infty$) is an order-reversing involution, hence a homeomorphism $\neg : (L, \sigma) \xrightarrow{\sim} (L, \omega)$ with $\omega = \neg_* \sigma$, and it conjugates scaling to its inverse, $\neg \circ D_k \circ \neg = D_{1/k}$;*

4. *the Lawson topology $\lambda = \sigma \vee \omega = \sigma \vee \neg_* \sigma$ is the compact Hausdorff Euclidean topology of $[0, \infty]$.*

Proof. Translation preserves order and directed suprema (with $T_s(\infty) = \infty$), so it is Scott-continuous, and its image $[s, \infty]$ is proper for $s > 0$. The way-below relation of the chain L is $a \ll b \iff a < b$ for $a > 0$ and $0 \ll 0$, which is exactly the displayed witnessing. Scaling is a bijective order map, hence a Scott homeomorphism. The reciprocal reverses order, so it is an order isomorphism $L \to L^{\mathrm{op}}$, i.e. a homeomorphism from the Scott topology to the Scott topology of the opposite order, which is the lower topology; $\neg_* \sigma = \omega$ follows by pushing forward the basic opens $(a, \infty]$ to $[0, 1/a)$. The last assertion is the classical fact that the join of the right and left half-open topologies on $[0, \infty]$ is the Euclidean topology [7, 2]. □

3.9 The affine hierarchy and the Scott–Lawson functorial square

The additive and multiplicative directions assemble into a four-level algebraic hierarchy, and the passage from the one-sided Scott side to the two-sided Lawson side is exactly the passage from the positive affine semigroup to the full projective group. Throughout, $\neg(z) = 1/z$ is the reciprocal of Proposition 3.9; it differs from the negative inversion $J(z) = -1/z$ of §3 by the reflection D_{-1}, via $J = D_{-1} \circ \neg = \neg \circ D_{-1}$, so that $\langle \mathrm{Aff}^+, \neg \rangle = \langle \mathrm{Aff}^+, J \rangle$ once D_{-1} is recovered (below).

Lemma 3.10 (Four-level affine hierarchy)**.** *Over $\mathbb{R}$, write*

$$\mathrm{Aff}^+ := \{a \mapsto ka + s : k > 0,\ s \geq 0\}, \qquad B_\infty = \mathrm{Stab}_{\mathrm{PGL}_2(\mathbb{R})}(\infty).$$

Then

$$\mathrm{Aff}^+ = B_\infty \cap \mathrm{Stab}([0,\infty]), \qquad \langle \mathrm{Aff}^+ \rangle = B_\infty^+ := \{a \mapsto ka + s : k > 0,\ s \in \mathbb{R}\},$$
$$B_\infty = B_\infty^+ \sqcup D_{-1} B_\infty^+, \qquad \mathrm{PGL}_2(\mathbb{R}) = \langle \mathrm{Aff}^+, \neg \rangle.$$

The reflection $D_{-1}(z) = -z$ is not in $\langle \mathrm{Aff}^+ \rangle$, but admits the explicit six-step decomposition

$$D_{-1} = T_1 \circ \neg \circ T_{-1} \circ \neg \circ T_1 \circ \neg \ \in\ \langle \mathrm{Aff}^+, \neg \rangle.$$

Hence the full affine group B_∞ already carries dual data borrowed through the second-slot division $\neg$; the purely one-sided Scott carrier is the positive affine semigroup Aff^+.

Proof. An affine map $a \mapsto ka + s$ preserves $[0,\infty]$ exactly when $k > 0$ and $s \geq 0$, giving the intersection formula. The group closure adds the inverses $T_{-s} = T_s^{-1}$ but keeps the diagonal positive, so it is B_∞^+; the full B_∞ is obtained by adjoining the reflection D_{-1}. The six-step decomposition is checked by applying the maps in the displayed order: $z \mapsto 1/z \mapsto 1/z + 1 \mapsto z/(1+z) \mapsto -1/(1+z) \mapsto -(1+z) \mapsto -z$. Finally $B_\infty^+ \cup \{D_{-1}\} \subseteq \langle \mathrm{Aff}^+, \neg \rangle$ forces $B_\infty \subseteq \langle \mathrm{Aff}^+, \neg \rangle$, and the Bruhat decomposition of §3 gives the whole of $\mathrm{PGL}_2(\mathbb{R})$. □

Theorem 3.11 (Scott–Lawson functorial square)**.** *On $L = [0,\infty]$ consider the projective restrictions $A(\tau) = C(L,\tau) \cap B_\infty$, $E(\tau) = C(L,\tau) \cap \mathrm{PGL}_2(\mathbb{R})$, and $H(\tau) = \mathrm{Homeo}(L,\tau) \cap \mathrm{PGL}_2(\mathbb{R})$.*

1. ***Homeomorphism level.*** *$H(\sigma) = \{D_k : k > 0\}$ and $H(\lambda) = \mathrm{Stab}([0,\infty]) = \langle H(\sigma), \neg \rangle$ is a proper subgroup of $\mathrm{PGL}_2(\mathbb{R})$, because the translations T_s are not onto and so are invisible to the homeomorphism level.*

2. ***Endomorphism level.*** *$A(\sigma) = \mathrm{Aff}^+$. On the generating set $S_0 = \{T_s (s \geq 0), D_k (k > 0)\}$, adjoining $\neg$ on the topological side (the patch topology $\tau \mapsto \tau \vee \neg_* \tau$) and adjoining $\neg$ on the group side (the closure $S \mapsto \langle S, \neg \rangle$) form a commutative square with the naturality identity*

$$\Phi(\sigma \vee \neg_* \sigma) = \langle \Phi(\sigma), \neg \rangle = \mathrm{PGL}_2(\mathbb{R}), \qquad \Phi(\sigma) = A(\sigma) = \mathrm{Aff}^+ .$$

3. ***No full Galois equivalence.*** *There is no single functor $\mathrm{Top}(L) \to \mathrm{Sub}(\mathrm{PGL}_2(\mathbb{R}))$ that is injective on $\{\sigma, \lambda\}$ and takes the values $(\mathrm{Aff}^+, \mathrm{PGL}_2)$: indeed $A(\sigma) = A(\lambda) = \mathrm{Aff}^+$ (collapse), $H(\lambda) \subsetneq \mathrm{PGL}_2$ (too small), and $E(\sigma) \ni \neg T_s \neg$ (too large, already containing the non-affine Möbius map $a \mapsto a/(1+sa)$).*

Proof. (1) A projective σ-homeomorphism is an order-preserving Möbius bijection of $[0,\infty]$, which must fix 0 and ∞, hence is $a \mapsto ka$ with $k > 0$; adjoining $\neg$ gives the set-wise stabilizer of $[0,\infty]$, which is the two classes $a \mapsto ka$ and $a \mapsto k/a$, a proper subgroup. (2) The affine Scott-continuous self-maps of L are exactly $a \mapsto ka + s$ with $k > 0, s \geq 0$ (order-preserving and mapping L into L), which is Aff^+; the commutativity is the observation that the patch topology $\sigma \vee \neg_* \sigma$ and the group closure $\langle \mathrm{Aff}^+, \neg \rangle$ both adjoin the same generator $\neg$ to the same base. (3) The three witnesses are displayed in the statement; $a \mapsto a/(1+sa) = \neg T_s \neg$ is order-preserving (Scott-continuous) but non-affine. □

3.10 Scott continuity of evaluation and the Hecke boundary

The projective evaluation map ρ of §3 is a monoid map, not an order-continuous one, and this fact pins down the scope boundary of the finite-history formalism.

Theorem 3.12 (Monotonicity obstruction)**.** *Let $\leq$ be any partial order on $\mathrm{PGL}_2(K)$. If $\rho : (\Gamma^*, \sqsubseteq) \to (\mathrm{PGL}_2(K), \leq)$ is monotone, then $\leq$ is the trivial preorder ($x \leq y$ for all x, y). Consequently ρ has no Scott-continuous extension to the history dcpo $\Gamma^{\leq\omega}$ valued in a non-degenerate dcpo. In the Lawson (product) topology on $\Gamma^{\leq\omega}$, the only candidate limit extension is the infinite matrix product*

$$\hat{\rho}(\gamma_\infty) = \lim_{n\to\infty} [M_n \cdots M_1],$$

whose convergence on the alternating history $\gamma_i = (\neg, T_{a_i})$ is exactly the convergence of the generalized continued fraction $[a_0; a_1, a_2, \dots]$.

Proof. Monotonicity of ρ under the prefix order, together with the anti-homomorphism $\rho(\gamma\delta) = \rho(\delta)\rho(\gamma)$ and the surjectivity of ρ (Theorem 3.4), gives $h \leq g\,h$ for all g, h. Taking $g = h^{-1}$ yields $h \leq 1$, and taking $h = 1$ yields $1 \leq g$; hence every element is comparable to 1 in both directions, i.e. the order is trivial. The Scott-continuity claim follows from the free-dcpo universal property, which extends a monotone map uniquely. The continued-fraction identification is the standard fact that an infinite product of Möbius maps evaluated at a base point is a generalized continued fraction. □

Proposition 3.13 (Hecke ends versus limit set)**.** *For the $q = 4$ Hecke sublanguage Hist_4 with alphabet $\Gamma_4 = \{\mathtt{t}_+, \mathtt{t}_-, \mathtt{j}\}$:*

1. *the completed history space $\Gamma_4^{\leq\omega}$ is compact Hausdorff in the Lawson topology, and its infinite boundary $\Gamma_4^\omega \cong 3^\omega$ is the standard Cantor space;*
2. *the projective group $G_4 = \langle T, J \mid J^2 = (JT)^4 = 1\rangle \cong C_2 * C_4$ is virtually free, and its space of ends $\mathrm{Ends}(G_4)$ (the boundary of the Bass–Serre tree) is a Cantor set, coded by the subshift of finite type of reduced words avoiding $\{JJ, (JT)^4, (TJ)^4\}$;*
3. *as a Fuchsian group, $G_4 = H(\sqrt{2})$ (parameter $\lambda_4 = 2\cos(\pi/4) = \sqrt{2} \leq 2$) is a first-kind lattice of finite covolume, whose geometric limit set is the whole real projective circle $\Lambda(G_4) = \mathbb{P}^1(\mathbb{R})$, not a Cantor set.*

Thus the history Cantor space codes the combinatorial space of ends, not the first-kind limit set; the latter belongs to the downstream function theory.

Proof. (1) is the product-topology description of the Lawson topology on the finite-alphabet word domain. (2) is the identification of G_4 with the free product $C_2 * C_4$ via $x = J$, $y = JT$, together with the classical description of the space of ends of a virtually free group as the boundary of its Bass–Serre tree. (3) is the standard fact that the Hecke triangle group of parameter $\lambda_q = 2\cos(\pi/q)$ is a lattice exactly for $\lambda_q \leq 2$. □

4 Analytical Pillar I: Differential Theory

The differential pillar turns the discrete histories of Section 2 into continuous flows and identifies the order defect of the minimal branching DAG with the contact curvature imported from Paper 0. Throughout this section μ, λ are the fixed real intensities of the regular AES of Section 7.

4.1 The smoothing Lie functor

Recall the arithmetic contact model $\mathcal{C} = \mathbb{R}^3_{(u,v,a)}$ with contact one-form

$$\alpha = da - \mu\, du - \lambda a\, dv,$$

horizontal fields

$$D_u = \partial_u + \mu\, \partial_a, \qquad D_v = \partial_v + \lambda a\, \partial_a,$$

and curvature bracket

$$[D_u, D_v] = \mu\lambda\, \partial_a$$

(`eq:arithmetic-contact-form`, `eq:horizontal-arithmetic-fields`, `thm:contact-curvature` of Paper 0). The additive and multiplicative generators of Section 2 are the exact closed-form flows of these two fields.

Lemma 4.1 (Generators are exact flows)**.** *The additive generator $A_h = T_{\mu h}$ and the positive multiplicative generator $M_k = D_{e^{\lambda k}}$ are the time-h and time-k flows of D_u and D_v acting on the assignment fiber:*

$$A_h(a) = \Phi^h_u(0,0,a)\big|_a = a + \mu h, \qquad M_k(a) = \Phi^k_v(0,0,a)\big|_a = e^{\lambda k} a,$$

where

$$\Phi^h_u(u,v,a) = (u+h, v, a+\mu h), \qquad \Phi^k_v(u,v,a) = (u, v+k, e^{\lambda k} a).$$

Proof. Direct substitution: the one-parameter groups of D_u and D_v are the displayed affine maps, read on the a-coordinate. □

Lemma 4.2 (Word-to-flow anti-homomorphism)**.** *The assignment F sending $A_h \mapsto \Phi^h_u$ and $M_k \mapsto \Phi^k_v$, and a chronological word to the composition of the corresponding flows, satisfies*

$$F(\gamma\delta) = F(\delta) \circ F(\gamma),$$

the same anti-homomorphism as the projective evaluation of Equation (8).

Proof. Immediate from the definition on generators and the free-monoid extension. □

4.2 Minimal DAG defects and contact curvature

The smallest nontrivial branching history is the two-element antichain $\{A, M\}$: addition and multiplication are incomparable operations. Its evaluation defect is exactly the contact curvature.

Theorem 4.3 (DAG defect equals contact curvature)**.** *Let $P = \{A, M\}$ be the two-element antichain (addition and multiplication incomparable). Its down-set lattice $\mathcal{O}(P) = B_2 = \{\varnothing, \{A\}, \{M\}, \{A, M\}\}$ is the four-element free distributive diamond, with Stone spectrum $\mathrm{Stone}(\mathcal{O}(P)) = \{A, M\}$, and its two maximal chains are the two spinal histories $\gamma_1 = (A_h, M_k)$ and $\gamma_2 = (M_k, A_h)$.*

1. ***Open and closed defects.*** *The open order defect is*

$$\nu_a(\gamma_1) - \nu_a(\gamma_2) = T_{\mathrm{open}}(h,k) = \mu h(e^{\lambda k} - 1),$$

and the four-step closed loop $(A_h, M_k, A_{-h}, M_{-k})$ has vertical drift

$$T_{\mathrm{closed}}(h,k) = \mu h(1 - e^{-\lambda k}) = e^{-\lambda k} T_{\mathrm{open}}(h,k).$$

2. ***Infinitesimal density equals the Lie bracket.***

$$\left.\frac{\partial^2 T_{\text{open}}}{\partial h\,\partial k}\right|_{0,0} = \left.\frac{\partial^2 T_{\text{closed}}}{\partial h\,\partial k}\right|_{0,0} = [D_u, D_v](a) = \mu\lambda.$$

3. ***Structural refinement.*** *The combinatorial diamond $\mathcal{O}(\{A, M\})$ and its Stone spectrum are flat (curvature zero); the curvature $\mu\lambda$ arises at the single step that lifts two directions, which commute at the charge level, to a non-commutative realization in the value space — the difference between the semi-direct rescaling law $D_k \circ T_s = T_{ks} \circ D_k$ and the naive swap $T_s \circ D_k$.*

Proof. (1) is the direct computation $\nu_a(A_h, M_k) = e^{\lambda k}(a+\mu h)$ versus $\nu_a(M_k, A_h) = e^{\lambda k}a+\mu h$, and the four-step substitution of Paper 0 (`eq:exact-open-contact-defect`, `eq:exact-closed-contact-drift`). (2) Differentiate: $\partial_h^2\partial_k^2\,\mu h(e^{\lambda k}-1) \to \mu\lambda$ and $\partial_h^2\partial_k^2\,\mu h(1-e^{-\lambda k}) \to \mu\lambda$, which is $[D_u, D_v](a) = \mu\lambda\,\partial_a(a) = \mu\lambda$ by `thm:contact-curvature`. (3) is the structural reading $D_k \circ T_s(a) = ka + ks = T_{ks} \circ D_k(a)$ versus $T_s \circ D_k(a) = ka + s$, whose difference is $s(k-1) = \mu h(e^{\lambda k}-1)$. □

4.3 Graded horizontal calculus

The horizontal differential of Paper 0 extends to a graded covariant differential with a twisted Leibniz law. Recall the scalar horizontal differential

$$\delta_H F = dF - (\partial_a F)\alpha = (D_u F)\,du + (D_v F)\,dv,$$

and the curvature operator on scalar fields

$$\mathcal{R}_H(F) = (D_u D_v F - D_v D_u F)\,du \wedge dv = \mu\lambda\,(\partial_a F)\,du \wedge dv$$

(`eq:scalar-horizontal-differential`, `prop:horizontal-differential-curvature` of Paper 0). In the natural arithmetic units $dA = \mu\,du$, $dM = \lambda\,dv$:

Definition 4.4 (Graded horizontal differential)**.** For a smooth section f of integer weight $n \in \mathbb{Z}$, define the *weight-n horizontal covariant differential*

$$\nabla^{(n)} f := \delta_H f - n\,f\,dM.$$

Proposition 4.5 (Graded Leibniz law and gauge rectification)**.** 1. *The assignment a has weight* 1 *and satisfies the propagation law*

$$\nabla^{(1)} a = \delta_H a - a\,dM = dA.$$

2. ***Graded Leibniz law.*** *For f of weight m and g of weight n,*

$$\nabla^{(m+n)}(fg) = (\nabla^{(m)} f)\,g + f\,(\nabla^{(n)} g).$$

3. ***Gauge rectification and twisted derivative.*** *Let $\widetilde{a} := e^{-M}a$ (weight 0); then $\delta_H\widetilde{a} = e^{-M}dA$. On a one-parameter curve, define the twisted product $(F \star_M G)(t) := e^{-M(t)}F(t)G(t)$ and the twisted derivative $D_M := \frac{d}{dt} - M'(t) = e^M \circ \frac{d}{dt} \circ e^{-M}$; then D_M is a strict derivation for $\star_M$:*

$$D_M(F \star_M G) = (D_M F) \star_M G + F \star_M (D_M G).$$

Proof. (1) $\delta_H a = \mu\,du + \lambda a\,dv$ along horizontal directions, so $\nabla^{(1)}a = \mu\,du + \lambda a\,dv - a\,\lambda\,dv = \mu\,du = dA$. (2) Since δ_H is a derivation on scalars, $\delta_H(fg) = (\delta_H f)g + f(\delta_H g)$, whence $\nabla^{(m+n)}(fg) = (\delta_H f - mf\,dM)g + f(\delta_H g - ng\,dM)$ as claimed. (3) $\delta_H\widetilde{a} = e^{-M}(\delta_H a - a\,dM) = e^{-M}dA$; and the conjugation $D_M = e^M \circ \frac{d}{dt} \circ e^{-M}$ is a derivation for the multiplicative structure induced by $\star_M$, checked directly. □

5 Analytical Pillar II: Integral Theory

The integral pillar has two layers. In one dimension it is the mixed additive–multiplicative integral whose two sides are Scott-continuous in the two directions of Section 3; in higher dimension it is the order defect as a group-valued 1-cocycle on the linear-extension graph, closed by the holographic Stokes theorem. Throughout, u, v are densities on $[a, b]$ and $M(t) = \int_a^t v$ is the multiplicative charge.

5.1 The one-dimensional mixed affine integral

Definition 5.1 (Mixed affine integral)**.** For $u, v \in L^1([a, b])$, define the *continuous mixed additive–multiplicative integral* in absolute-time parametrization,

$$\mathfrak{M}_a^b[u; v] := \int_a^b u(x)\, \exp\Big(\int_x^b v(s)\, ds\Big)\, dx,$$

and, in the right-endpoint sliding-window parametrization, the one-sided Volterra convolution

$$\mathfrak{A}(w) := \int_0^w u(x)\, K(w - x)\, dx, \qquad K(r) = \exp\Big(\int_0^r v(y)\, dy\Big).$$

Proposition 5.2 (Mixed-integral equivalences)**.** *1.* ***Affine transport and splitting.*** *$\mathfrak{M}_a^b[u; v]$ is the inhomogeneous propagation term of the affine transport equation $w'(t) = v(t)w(t) + u(t)$:*

$$w(b) = e^{\int_a^b v} w(a) + \mathfrak{M}_a^b[u; v],$$

and satisfies the splitting law and the right-endpoint derivative

$$\mathfrak{M}_a^c[u; v] = e^{\int_b^c v} \mathfrak{M}_a^b[u; v] + \mathfrak{M}_b^c[u; v], \qquad \frac{d}{db}\mathfrak{M}_a^b[u; v] = u(b) + v(b)\, \mathfrak{M}_a^b[u; v].$$

2. ***Window derivative and integration by parts.*** *For $u \in C^1([0, w])$, $v \in C([0, w])$,*

$$\mathfrak{A}'(w) = u(0)K(w) + \int_0^w u'(x)K(w - x)\, dx = u(w) + \int_0^w u(x)\, v(w - x)K(w - x)\, dx.$$

3. ***Reverse ACS line integral.*** *On the accumulative commutative space $\mathcal{A} = \mathbb{R}^2_{(A,M)}$,*

$$\mathfrak{M}_a^b[u; v] = \int_{C^{\mathrm{rev}}} \eta, \qquad \eta = e^M dA,$$

where C^{rev} is the ACS path generated by the reversed density.

4. ***Twisted Rota–Baxter identity.*** *With $(I_M f)(t) = e^{M(t)} \int_{t_0}^t e^{-M(s)} f(s)\, ds = (e^M \circ J \circ e^{-M})f$ and the twisted product $\star_M$ of Proposition 4.5,*

$$I_M(f) \star_M I_M(g) = I_M(I_M(f) \star_M g + f \star_M I_M(g)).$$

Proof. (1) Differentiate $e^{-\int_a^t v} w(t)$ to isolate the inhomogeneous term; the splitting law is the semigroup property of the fundamental solution, and the endpoint derivative is the transport equation itself. (2) Differentiate the convolution under the sign and integrate by parts. (3) Substitute A, M and observe that the density $\eta = e^M dA$ integrates along the reversed path to the displayed form. (4) is a direct computation using the twisted derivation law of Proposition 4.5. □

5.2 Dual adjoint pairing and the Lawson two-sided functional

The two directions of the mixed integral are not symmetric: the additive source is the Scott (from-below) side and the multiplicative density is the lower (dual) side. Their pairing is the functional form of the Lawson topology.

Theorem 5.3 (Lawson dual adjoint pairing)**.** *1.* ***Dual connection.*** *On the weight-(-1) dual line, define $\nabla^{(-1)}p := \delta_H p + p\,dM$. Then*

$$\delta_H(pa) = (\nabla^{(-1)}p)a + p\,(\nabla^{(1)}a),$$

and when p satisfies the homogeneous backward-parallel equation $\nabla^{(-1)}p = 0$, i.e. $-p'(t) = v(t)p(t)$ with $p(b) = 1$, one has $\delta_H(pa) = p\,dA$, so that

$$\mathfrak{M}_a^b[u; v] = \int_a^b p(t)\,u(t)\,dt.$$

2. ***Backward adjoint equals the reciprocal image.*** *With the forward scale $K(t) = e^{M(t)}$ and the backward adjoint $p(t) = \exp\big(\int_t^b v(s)\,ds\big)$,*

$$p(t) = e^{M(b)-M(t)} = K(b) \cdot \neg(K(t)), \qquad \neg(a) = \frac{1}{a}.$$

3. ***Bilateral Scott continuity.*** *For nonnegative densities $u, v \geq 0$, the functional $u \mapsto \mathfrak{M}[u; v]$ and $v \mapsto \mathfrak{M}[u; v]$ are each monotone and preserve directed suprema, hence are separately Scott-continuous; the pairing $\int_a^b u\,p$ is the functional-level realization of the Lawson topology $\lambda = \sigma \vee \neg_*\sigma$ of Proposition 3.9.*

Proof. (1) is the graded Leibniz law of Proposition 4.5 at weights $-1, 1$; solving $\nabla^{(-1)}p = 0$ gives $p(t) = e^{M(b)-M(t)}$. (2) is the displayed computation. (3) monotone convergence gives the directed-sup preservation in each argument; the forward scale K and the backward adjoint $p = \neg(K)$ carry the Scott and the lower side respectively. □

5.3 The order defect as a group-valued 1-cocycle

For a general dependency poset P the order defect between two linear extensions is additive along adjacent swaps, and closed by the Jacobi identity.

Theorem 5.4 (Defect group 1-cocycle)**.** *Let P be a finite dependency poset and $G(P)$ its linear-extension graph (vertices are linear extensions γ, edges are adjacent transpositions of incomparable elements).*

1. ***Exact telescoping.*** *For a path $\gamma_0 \to \gamma_1 \to \cdots \to \gamma_m$ in $G(P)$,*

$$\rho(\gamma_m)\rho(\gamma_0)^{-1} = \delta(\gamma_{m-1} \to \gamma_m) \cdots \delta(\gamma_0 \to \gamma_1), \qquad \delta(\gamma \to \gamma') := \rho(\gamma')\rho(\gamma)^{-1},$$

and the scalar defect $\nu_a(\gamma_0) - \nu_a(\gamma_m)$ is the corresponding telescoping sum. Both depend only on the endpoints.

2. ***Suffix-conjugated commutator.*** *In a history $\gamma = UxyW$ (U executed first, W last), swapping adjacent incomparable x, y to $\gamma' = UyxW$ gives*

$$\delta(UxyW \to UyxW) = \rho(W)\,[\rho(x), \rho(y)]\,\rho(W)^{-1}, \qquad [g, h] := ghg^{-1}h^{-1}.$$

3. ***BCH expansion and Jacobi closure.*** *With infinitesimal generators* e^{hX}, e^{kY}, e^{lZ},

$$[e^{hX}, e^{kY}] = \exp\Big(hk[X,Y] + \tfrac{h^2k}{2}[X,[X,Y]] + \tfrac{hk^2}{2}[Y,[X,Y]] + \cdots\Big),$$

and the two three-step paths around the hexagon of a three-element antichain $\{X, Y, Z\}$ *differ, to third order, by exactly the Jacobi identity* $[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$ *(the discrete Bianchi identity* $d\Omega = 0$*).*

Proof. (1) is the telescoping product; (2) is the direct computation $\rho(UyxW)\rho(UxyW)^{-1} = \rho(W)\rho(x)\rho(y)\rho(x)^{-1}\rho(y)^{-1}\rho(W)^{-1}$; (3) is the Campbell–Baker–Hausdorff formula and the vanishing of its third-order mixed term by the Jacobi identity. □

5.4 Holographic Stokes theorem

The two-dimensional order defect is a boundary quantity: the bulk carries an exact curvature form, and Stokes pushes its integral to the boundary holonomy.

Proposition 5.5 (Holographic Stokes, minimal case)**.** *In the contact realization of the minimal branching DAG* $\{A, M\}$*, the curvature two-form*

$$d\alpha = -\lambda\, da \wedge dv = d(-\lambda a\, dv)$$

is nonzero but exact. For the charge rectangle $\Sigma = [0,h] \times [0,k]$*, Stokes gives*

$$T = \int_\Sigma d\alpha = \oint_{\partial\Sigma} (-\lambda a\, dv),$$

i.e. the order defect is the boundary holonomy of the potential one-form $-\lambda a\, dv$ *around the closed loop* $\partial\Sigma = (A_h, M_k, A_{-h}, M_{-k})$*.*

Proof. $d\alpha = -\lambda\, da \wedge dv$ is exact with primitive $-\lambda a\, dv$; Stokes identifies the area integral with the boundary line integral, and the boundary integral is the closed-loop drift of Theorem 4.3. □

Remark 5.6 (The ideal surface as the true limit). On the completion axis, the finite DAGs (compact elements) form a flat "bulk" and the infinite histories Γ^ω form an "ideal surface". The Scott topology keeps the infinite histories as genuine limits, so the curvature exists as a surface holonomy, whereas the Alexandrov topology would reify them as isolated points and erase the bulk–surface distinction. The macroscopic discrete Stokes integral over an arbitrary dependency poset P and its spectral docking with the down-set lattice of Theorem 2.17 are open (see OQ-095.3, OQ-095.4). This remark is a structural proposal, not a theorem.

6 Analytical Pillar III: Series Theory

The series pillar expands the history evaluation into the free Lie algebra: the formal logarithm of a word is a Magnus series whose arity-k terms are carried by the $(k-1)$-cells of the down-set lattice of Theorem 2.17.

6.1 The Magnus series of a history

Definition 6.1 (History Magnus series)**.** For a chronological history $\gamma = (e^{X_1}, \dots, e^{X_n})$ of infinitesimal generators, the formal logarithm of its evaluation $\rho(\gamma) = e^{X_n} \cdots e^{X_1}$ is the *Magnus series* in the completed free Lie algebra $\mathrm{Lie}(X_1, \dots, X_n)$:

$$\Omega(\gamma) = \log(e^{X_n} \cdots e^{X_1}) = \underbrace{\sum_{i=1}^{n} X_i}_{\Omega_1} + \underbrace{\tfrac{1}{2} \sum_{i<j} [X_j, X_i]}_{\Omega_2} + \underbrace{\tfrac{1}{12} \sum_{i,j,k} \big(2[X_i, [X_j, X_k]] + \cdots\big) + \cdots}_{\Omega_3}$$

The three leading orders have the following meaning: Ω_1 is the abelian charge (the sum of the generators), Ω_2 is the commutator (the curvature of Section 4), and Ω_3 and higher are the nested brackets. The exact coefficients are recorded, together with their symbolic verification, in Appendix C.

6.2 Arity–cell correspondence and DAG completeness

Decompose the brackets of $\Omega(\gamma)$ by the number of *distinct* generators they contain (the arity) versus the repeated-generator self-nesting.

Proposition 6.2 (Arity–cell correspondence, low order)**.**

1. ***Low orders (verified).*** *For the Magnus series of Definition 6.1:*

   - arity $= 1$ *(the generators X_i) $\leftrightarrow$ the 0-cells of $\mathcal{O}(P)$ (the scalar charges);*
   - arity $= 2$ *(the commutators $[X_i, X_j]$) $\leftrightarrow$ the 1-cells of $\mathcal{O}(P)$ (the two-element antichains, i.e. the diamonds B_2), whose coefficient is the contact curvature $\mu\lambda$;*
   - arity $= 3$ *(the nested brackets $[X,[Y,Z]]$, $[Y,[Z,X]]$, $[Z,[X,Y]]$) $\leftrightarrow$ the 2-cells of $\mathcal{O}(P)$ (the three-element antichains). In the order-three free nilpotent matrix model $X = E_{12}$, $Y = E_{23}$, $Z = E_{34}$ the coefficient of the E_{14}-term of $\log(e^{hX} e^{kY} e^{lZ})$ is $\frac{1}{3}hkl\,[X,[Y,Z]]$ (Appendix C).*

   Repeated-generator terms such as $[X,[X,Y]]$ do not correspond to a new antichain cell; they are the higher-order self-corrections of the single flow e^{hX}.

2. ***DAG completeness.*** *The two-dimensional contact distribution $\mathcal{D} = \mathrm{span}\{D_u, D_v\}$ has only two generators, so its free Lie algebra has* arity ≤ 2*: every bracket of depth three or more is a repeated-generator self-nesting. Only on a general branching DAG with n mutually incomparable operations does $\mathrm{Lie}(X_1, \dots, X_n)$ contain arity-$3, \dots, n$ brackets, encoding the higher-dimensional cells of $\mathcal{O}(P)$.*

3. ***Open boundary.*** *The general arity-k localization theorem (that the arity-k terms are carried exactly by the k-element antichains) and the full DAG Lie realization functor $F : \mathrm{FinDAG} \to \mathrm{Lie}(g_1, \dots, g_n)$ are open (OQ-095.1, OQ-095.2); beyond order three they are a structural proposal, not a theorem.*

Proof. The low-order correspondence is the computation of Appendix C; the DAG-completeness observation is the rank of the free Lie algebra on two versus on n generators; the open boundary is stated explicitly and not proved here. □

6.3 Bulk–boundary reduction of the series

Proposition 6.3 (Bulk–boundary series reduction)**.** *The Magnus series of the bulk undergoes three simplifications on projection to the surface:*

1. ***Degree descent.*** *The two-form curvature carrier $d\alpha$ (corresponding to the arity-2 brackets) descends to the one-form potential $-\lambda a\, dv$ on the surface (Stokes of Proposition 5.5);*
2. ***Cocycle reduction.*** *By the closure of Theorem 5.4, the interior-cell higher-order terms cancel in pairs around the closed boundary, leaving only the cohomology class;*
3. ***Canonical form.*** *The Magnus series depends on the ordering of the generators and the Hall/Lyndon basis (non-canonical), while its evaluation at the ideal boundary Γ^ω is the infinite Möbius product, i.e. the continued fraction, which is canonical (unique partial quotients).*

Remark 6.4 (Status of the reduction). Items (1) and (2) are proved (the exactness of $d\alpha$ and the 1-cocycle closure of Sections 4 and 5). Item (3), the "canonical form" reading, is a structural proposal: the continued fraction is the canonical infinite-product representation, but the claim that it is the surface reduction of the Magnus series is a reading, not a theorem.

6.4 PBW abelianization and the difference substitution

The abelianization of the Magnus series is exactly the cumulative sum, which is the algebraic form of the difference substitution of automated inequality proving.

Theorem 6.5 (PBW abelianization equals the difference substitution)**.** *Let $\mathfrak{L} = \mathrm{Lie}(X_1, \ldots, X_n)$. By the Poincaré–Birkhoff–Witt theorem, $\mathrm{gr}(U(\mathfrak{L})) \cong S(\mathfrak{L})$, the symmetric algebra. Setting all brackets to zero (abelianization, i.e. the contact defect $\mu\lambda = 0$):*

1. *the Lyndon/Hall basis of $\mathfrak{L}$ degenerates to the commutative monomial basis $d_1^{\alpha_1} \cdots d_n^{\alpha_n}$ of $S(\mathfrak{L})$;*
2. *the non-commutative Magnus series $\log(e^{X_1} \cdots e^{X_j})$ degenerates to the abelian cumulative sum $x_j = d_1 + \cdots + d_j$;*
3. *this cumulative sum and the monomial expansion are the algebraic structure of the* difference substitution *used in automated inequality proving: on the non-commutative side one extracts the Magnus defect coefficients, while on the commutative side one tests the nonnegativity of the monomial coefficients over all $n!$ orderings.*

Proof. (1) is the PBW theorem; (2) is the BCH formula with all brackets set to zero, $e^X e^Y = e^{X+Y}$; (3) is the structural identification of the cumulative sum with the difference substitution, stated as a correspondence rather than a claim of equivalence of the two algorithms. □

Remark 6.6 (Wu–Zhang nondegeneracy correspondence). The pseudodivision of the Wu elimination method (multiplying instead of dividing, pushing the singularity to the initial $\mathrm{lc} \neq 0$ boundary) and the weak-nondegeneracy improvement of the area method correspond structurally to replacing, in AEG, the naive swap $T_s \circ D_k$ (with defect $\mu\lambda$) by the semi-direct rescaling law $D_k \circ T_s = T_{ks} \circ D_k$ (no defect). The two degenerations differ: $\mathrm{lc} = 0$ is non-invertibility of the multiplicative side, while $\mu\lambda = 0$ is integrability of the horizontal distribution; and the carriers differ (the polynomial ring versus the affine flow). This correspondence is a structural proposal, not a theorem.

7 The Regular AES Interface

This short section restates the one geometric definition on which the zero theorems of Section 8 depend. It is imported from Paper 0, where the definition, the frame equivalence, the models, the flow, the commutativization, and the contact structure are developed and proved; the statements below are repeated here only so that this paper can be read on its own, and they carry the same labels and hypotheses as their source.

Definition 7.1 (Regular arithmetic expression space, imported)**.** A *regular arithmetic expression space*, or regular AES, is a tuple

$$\mathfrak{E} = (M, g, a; \mu, \lambda)$$

such that M is an oriented smooth surface, possibly with boundary, g is a smooth Riemannian metric, $a \in C^\infty(M, \mathbb{R})$, $\mu \in \mathbb{R} \setminus \{0\}$, $\lambda \in \mathbb{R}$, and

$$|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2 \tag{12}$$

everywhere on M. The function a is called the *assignment function*.

The metric is part of the structure; (12) is an eikonal equation on a specified Riemannian surface, not a metric-free identity. Paper 0 defines the arithmetic motions of such a space and proves that they realize addition, subtraction, multiplication, and division by a constant as motions of M.

Proposition 7.2 (Canonical arithmetic frame, imported)**.** *For an oriented Riemannian surface (M, g), a smooth function a, and constants $\mu \neq 0$ and λ, the following are equivalent.*

1. *The eikonal identity (12) holds.*
2. *There is a unique positively oriented orthonormal frame (X_u, X_v) satisfying*

$$X_u a = \mu, \qquad X_v a = \lambda a. \tag{13}$$

When these conditions hold, the frame is

$$X_u = \frac{\mu}{q} E_a - \frac{\lambda a}{q} J E_a, \qquad X_v = \frac{\lambda a}{q} E_a + \frac{\mu}{q} J E_a, \qquad E_a = \frac{\nabla a}{q}, \qquad q = \sqrt{\mu^2 + \lambda^2 a^2}, \tag{14}$$

and it satisfies

$$[X_u, X_v] a = \mu\lambda. \tag{15}$$

The proof is given in Paper 0. We use only the following immediate consequences, which is why the statement is repeated rather than cited.

1. $\mu \neq 0$ makes q everywhere positive, so $|\nabla a|$ never vanishes. This is the source of the regularity of zero sets below.
2. The bracket identity (15) shows that, when $\lambda \neq 0$, the two arithmetic directions cannot be realized as a coordinate frame.

No other object of Paper 0 is repeated in this paper. In particular, the basic hyperbolic model $\mathfrak{E}_0$, the punctured model $\mathfrak{E}_1$, the continuous arithmetic motion, the affine cocycles, the accumulative commutative space, tearing, and the contact structure are cited where needed and never reproved here.

8 Holes and Zero Geometry

The eikonal equation imposes an immediate restriction on the zero set of a regular assignment: its gradient cannot vanish there. This separates regular zero geometry from isolated zeros, crossings, and the parameter-dependent singular phenomena reserved for later work.

8.1 A taxonomy of holes and boundaries

Five notions of "hole" occur across the series and must not be conflated.

Definition 8.1 (Fivefold hole taxonomy)**.** 1. **Syntactic one-hole placeholder.** The formal slot z of a one-hole context $C^{(\varepsilon)}_{\omega,c}[z]$ of Section 2.

2. **Computational port / wire hole.** A typed input or output occurrence of a strict finite directed acyclic network, awaiting a substitution binding, with a strict partial definition domain such as discard(1/0).

3. **Topological surface hole / seam circle.** A boundary circle of the side surface of a thickened body, or the seam circle $C_e = \partial D_e$ exposed when a thickened body is cut along an internal interface disk (Section 9).

4. **Essential geometric puncture / singular set.** A closed nowhere-dense subset $S \subset \overline{M}$ on which the regular AES data $(g, a; \mu, \lambda)$ admit no smooth nondegenerate local extension.

5. **Combinatorial interaction site.** A channel of a multi-component coupled system carrying the two-way exchange and the frame/phase monodromy twist (as in the K_3 edges and $\mathbb{Z}_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_{12}$ twist of Paper 0).

Convention 8.2 (Four boundaries)**.** 1. **Computational frontier.** The directed wire occurrences cut between a down-set $U \in \mathcal{O}(P)$ and the uncompleted events.

2. **Geometric boundary.** The boundary of a thickened body, $\partial H_G = S_G \cup \bigcup_{D \in E_G} D$.

3. **Integration boundary.** The oriented one-chain $\partial\Sigma$ of the Stokes formula $\int_\Sigma d\alpha = \int_{\partial\Sigma} \beta$.

4. **Completion boundary.** The infinite histories Γ^ω of the ideal completion of Proposition 2.14.

8.2 Regular zero loci

For an assignment function a, write

$$Z(a) = a^{-1}(0).$$

Theorem 8.3 (Regular zero locus)**.** *Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES.*

1. *If M has no boundary, then $Z(a)$ is a smooth embedded codimension-one submanifold of M.*

2. *If M has boundary, then $Z(a) \cap \operatorname{int} M$ is a smooth embedded codimension-one submanifold of $\operatorname{int} M$.*

In particular, for a surface the indicated zero locus is a disjoint union of smooth one-dimensional components. No assertion is made here about finiteness, properness, compactness, or behavior at ∂M.

Proof. At every $p \in Z(a)$, the defining eikonal identity gives

$$|\nabla a(p)|_g^2 = \mu^2.$$

Because $\mu \neq 0$, one has $da_p \neq 0$. Thus 0 is a regular value of a when M is boundaryless, and of $a|_{\operatorname{int} M}$ for interior zeros when a boundary is present. The regular-value theorem gives the stated embedded submanifolds. □

Corollary 8.4 (Local rigidity of regular zeros)**.** *Within the interior of a regular two-dimensional AES, the zero locus has no isolated point, crossing, branch point, or endpoint. None of these configurations can occur as the germ of $Z(a)$ at an interior point.*

Proof. By Theorem 8.3, a neighborhood of every interior zero admits smooth coordinates in which $a^{-1}(0)$ is a coordinate line. Such a germ has none of the listed configurations. □

This conclusion is local. Without additional global hypotheses, it does not classify the number or topology of the components. In the basic hyperbolic model, for example, $Z(a) = \{x = 0, y > 0\}$. Completeness and the absence of a boundary strengthen the conclusion substantially.

8.3 Complete regular splitting and a global no-go theorem

Theorem 8.5 (Complete regular AES splitting)**.** *Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES for which M is connected, has no boundary, and is complete as a Riemannian manifold. Define*

$$\mathcal{R}(t) = \int_0^t \frac{ds}{\sqrt{\mu^2 + \lambda^2 s^2}}, \qquad r = \mathcal{R} \circ a. \tag{16}$$

Then $|\nabla r|_g = 1$, the vector field $X = \nabla r$ has a global flow $\Phi : \mathbb{R} \times M \to M$, and

$$F : Z(a) \times \mathbb{R} \longrightarrow M, \qquad F(q, t) = \Phi_t(q), \tag{17}$$

is a diffeomorphism. In particular, a is surjective, $Z(a)$ is nonempty and connected, and M is noncompact.

The conclusion is a smooth product decomposition. It does not assert that F is an isometry from a Riemannian product.

Proof. Put $q(a) = \sqrt{\mu^2 + \lambda^2 a^2}$. Since $\mu \neq 0$, this is everywhere positive. The chain rule and the regular-AES equation give

$$\nabla r = \frac{\nabla a}{q(a)}, \qquad |\nabla r|_g^2 = \frac{|\nabla a|_g^2}{q(a)^2} = 1.$$

Thus $X = \nabla r$ is a smooth bounded vector field. To see that its flow is complete, let an integral curve be defined on a finite time interval. Its speed is one, so over every bounded time interval it remains in a closed metric ball. By Hopf–Rinow, closed bounded balls in the complete connected Riemannian manifold M are compact. The standard local existence theorem for smooth vector fields then extends the integral curve past every finite endpoint. Hence Φ_t exists for all $t \in \mathbb{R}$.

Along this flow,

$$\frac{d}{dt} r(\Phi_t(p)) = dr(X) = |\nabla r|_g^2 = 1, \qquad r(\Phi_t(p)) = r(p) + t. \tag{18}$$

It follows that every orbit meets $r^{-1}(0) = a^{-1}(0)$ exactly once, namely at $\Phi_{-r(p)}(p)$. Therefore the inverse of (17) is

$$p \longmapsto \big(\Phi_{-r(p)}(p), r(p)\big).$$

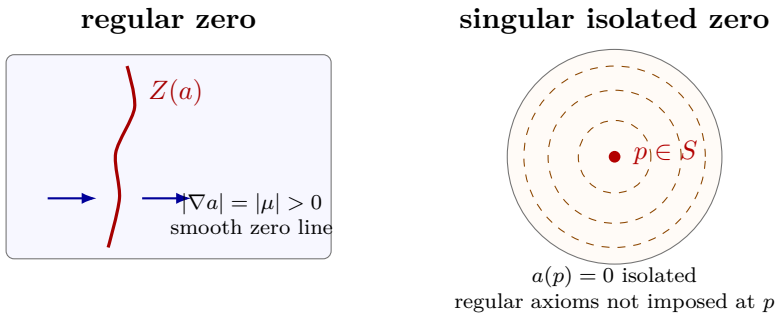


Figure 5: Regular and singular zero geometry. In a regular AES with $\mu \neq 0$, the assignment crosses zero with nonvanishing gradient, so the interior zero germ is a smooth curve. The radial disc model instead has an isolated zero at its declared singular point, where the assignment is not C^1 and the regular structure has no local extension.

Both maps are smooth, so F is a diffeomorphism. Equation (18) also shows that $r(M) = \mathbb{R}$. The function $\mathcal{R} : \mathbb{R} \to \mathbb{R}$ is a strictly increasing diffeomorphism (linear when $\lambda = 0$, and an inverse-hyperbolic-sine rescaling when $\lambda \neq 0$); hence a is surjective. Finally, $Z(a) \times \mathbb{R} \cong M$ is connected only if $Z(a)$ is connected, and the displayed product cannot be compact. □

Remark 8.6 (Why the global hypotheses matter). If a boundary is present, the unit vector field ∇r may reach it in finite time. If the metric is incomplete, an integral curve may escape the manifold in finite time. In either case the global flow argument fails, so the theorem must not be applied to domains with boundary or to incomplete regular models without a separate extension analysis.

Theorem 8.5 is a global obstruction, not merely a local regular-value statement. A connected complete boundaryless regular scalar AES cannot have several zero components, let alone a branching or four-valent zero network. Consequently, a Hecke cellulation used as a zero graph must enter through singular points, a failure of completeness or boundarylessness, or data richer than one regular scalar assignment. Paper I records this no-go boundary; the corresponding singular constructions belong to Paper III.

Figure 5 summarizes the categorical distinction. It illustrates the local conclusions already proved; it does not classify singular zero germs.

8.4 Singular arithmetic expression spaces

The preceding theorem makes a singular framework unavoidable if one wishes to admit isolated or branching zeros.

Definition 8.7 (Singular arithmetic expression space)**.** A *singular arithmetic expression space* is a tuple

$$\mathfrak{E}_{\text{sing}} = (M, S, g, a; \mu, \lambda)$$

with the following data:

1. M is a nonempty oriented smooth surface, possibly with boundary, and $S \subset M$ is a closed nowhere-dense singular set;
2. g is a smooth Riemannian metric on $M \setminus S$;
3. $a : M \to \mathbb{R}$ is a specified function whose restriction to $M \setminus S$ is smooth;

4. $\mu \in \mathbb{R} \setminus \{0\}$, $\lambda \in \mathbb{R}$, and
$$(M \setminus S, g, a|_{M\setminus S}; \mu, \lambda)$$
is a regular AES.

5. Every $p \in S$ is locally essential: there is no neighborhood U of p on which $g|_{U\setminus S}$ and $a|_{U\setminus S}$ admit smooth extensions $\widetilde{g}$ and $\widetilde{a}$, with $\widetilde{g}$ nondegenerate, such that $(U, \widetilde{g}, \widetilde{a}; \mu, \lambda)$ is a regular AES.

The metric or assignment may possess weaker extensions across S; their precise regularity and local behavior must be checked in each example.

The set S need not consist entirely of zeros. Nowhere density keeps the regular locus nonempty and dense, while local essentiality prevents an arbitrary enlargement of S by points at which the regular structure extends. For a singular AES define

$$Z_{\mathrm{reg}}(a) = Z(a) \cap (M \setminus S), \qquad Z_{\mathrm{sing}}(a) = Z(a) \cap S. \tag{19}$$

Every point of $Z_{\mathrm{reg}}(a)$ is a regular zero by Theorem 8.3; a point of $Z_{\mathrm{sing}}(a)$ is called singular because the regular structure is not asserted there, even when some of the underlying data happen to extend smoothly.

8.5 A verified isolated-zero singular model

Let
$$\mathbb{D} = \{(\xi, \eta) \in \mathbb{R}^2 : \xi^2 + \eta^2 < 1\}, \qquad \rho = \sqrt{\xi^2 + \eta^2},$$
and fix $\mu, \lambda > 0$. Equip the disc with

$$g_{\mathbb{D}} = \frac{4}{\lambda^2(1-\rho^2)^2}(d\xi^2 + d\eta^2) \tag{20}$$

and define

$$a_{\mathbb{D}}(\xi, \eta) = \begin{cases} \dfrac{2\mu}{\lambda}\dfrac{\rho}{1-\rho^2}, & (\xi, \eta) \neq (0, 0), \\ 0, & (\xi, \eta) = (0, 0). \end{cases} \tag{21}$$

Proposition 8.8 (Classification of the isolated-zero disc)**.** *The tuple*

$$(\mathbb{D}, \{0\}, g_{\mathbb{D}}, a_{\mathbb{D}}; \mu, \lambda)$$

is a singular AES with the following properties.

1. *The metric $g_{\mathbb{D}}$ extends smoothly and nondegenerately across the center.*

2. *The assignment $a_{\mathbb{D}}$ is smooth on $\mathbb{D} \setminus \{0\}$ but is not C^1 at the center.*

3. *The punctured disc is a regular AES:*
$$|\nabla a_{\mathbb{D}}|^2_{g_{\mathbb{D}}} = \mu^2 + \lambda^2 a_{\mathbb{D}}^2.$$

4. *Its zero decomposition is*
$$Z_{\mathrm{reg}}(a_{\mathbb{D}}) = \varnothing, \qquad Z_{\mathrm{sing}}(a_{\mathbb{D}}) = \{0\}.$$

Proof. Formula (20) is written in Cartesian coordinates. Its conformal factor is a positive smooth function on all of $\mathbb{D}$, proving the first claim.

Away from the origin, ρ and hence $a_{\mathbb{D}}$ are smooth. On the ξ-axis, however,

$$\frac{a_{\mathbb{D}}(t,0) - a_{\mathbb{D}}(0,0)}{t} = \frac{2\mu}{\lambda} \frac{|t|}{t(1-t^2)}.$$

The one-sided limits are $2\mu/\lambda$ and $-2\mu/\lambda$. Thus even the partial derivative in the ξ-direction does not exist at the center, so the assignment is not C^1 there. Moreover, the punctured assignment tends to zero at the center, so its continuous extension is unique. The same calculation therefore rules out every smooth local extension agreeing with the punctured assignment and proves that the declared singular point is locally essential.

On the punctured disc the assignment is radial, and

$$\frac{da_{\mathbb{D}}}{d\rho} = \frac{2\mu}{\lambda} \frac{1+\rho^2}{(1-\rho^2)^2}.$$

The inverse radial metric coefficient is

$$g_{\mathbb{D}}^{\rho\rho} = \frac{\lambda^2(1-\rho^2)^2}{4}.$$

Consequently,

$$\begin{aligned} |\nabla a_{\mathbb{D}}|^2_{g_{\mathbb{D}}} &= g_{\mathbb{D}}^{\rho\rho} \left(\frac{da_{\mathbb{D}}}{d\rho} \right)^2 \\ &= \mu^2 \frac{(1+\rho^2)^2}{(1-\rho^2)^2} \\ &= \mu^2 + \lambda^2 \left(\frac{2\mu}{\lambda} \frac{\rho}{1-\rho^2} \right)^2, \end{aligned}$$

which is the regular-AES equation on $\mathbb{D} \setminus \{0\}$. Finally, $a_{\mathbb{D}} > 0$ for $0 < \rho < 1$, while its declared value at the center is zero. The claimed zero decomposition follows. □

If $s = (2/\lambda) \operatorname{arctanh} \rho$ denotes hyperbolic distance from the center, then

$$a_{\mathbb{D}} = \frac{\mu}{\lambda} \sinh(\lambda s).$$

This identity describes radial propagation on the punctured regular part; it does not repair the failure of differentiability at the center. The metric is regular there, while the assignment is singular.

8.6 Two essential puncture mechanisms

The isolated-zero disc is one of two verified mechanisms by which an essential puncture arises; they differ in which datum degenerates.

Theorem 8.9 (Two essential puncture mechanisms)**.** *1.* ***Type 1: metric extends, assignment is not*** C^1***.*** *The disc model* $\mathfrak{E}_1$ *of Proposition 8.8: the Poincaré metric extends smoothly and nondegenerately across the center, while the assignment is continuous but not* C^1 *there, with one-sided directional limits* $\pm 2\mu/\lambda$*. Hence* $S = \{0\}$ *is essential,* $Z_{\text{reg}} = \varnothing$*,* $Z_{\text{sing}} = \{0\}$*, and the punctured disc is metrically incomplete (the puncture is at finite distance).*

2. ***Type 2: assignment extends, conformal background degenerates.*** *When the assignment a extends smoothly to $\widetilde{a}$ on $\overline{M}$, and the metric is conformal to a smooth nondegenerate background g_0 on $U \setminus S$, the essential puncture set is exactly the critical locus:*

$$S = \mathrm{Crit}(\widetilde{a}) = \{p : d\widetilde{a}(p) = 0\}, \qquad g = \frac{|d\widetilde{a}|^2_{g_0}}{\mu^2 + \lambda^2 \widetilde{a}^2}\, g_0.$$

In the four-circle polynomial model the tangent cone at the origin is the double coordinate-axes union $(xy)^2 = 0$, and the zero locus is four circles with eight branches meeting at the puncture, the conformal factor degenerating like $O(|x|^{10})$ so that the puncture is at finite distance (`thm:p0-punctures-are-critical`, `thm:p0-four-circle-model` of Paper 0).

Remark 8.10 (The criticality hypothesis). The inclusion $\mathrm{Crit}(\widetilde{a}) \subseteq S$ always holds when $\widetilde{a}$ extends smoothly; the reverse inclusion $S \subseteq \mathrm{Crit}(\widetilde{a})$ requires the stated conformal-background hypothesis. Without it, a counterexample with $da \neq 0$ yet singular metric at a puncture is recorded (C-10 of the notes reserve); the two inclusions must not be conflated.

8.7 The puncture holonomy

The essential puncture of Definition 8.1 has an analytic residue alongside its topological essentiality, parallel to the seam-circle holonomy of Section 9. On the punctured regular part the surface potential

$$\beta = (-\lambda a\, dv)\big|_{M \setminus S}$$

of Definition 9.6 is available.

Proposition 8.11 (Puncture holonomy)**.** *Let γ be a small loop in $M \setminus S$ encircling an essential puncture $p \in S$. The loop integral*

$$\oint_\gamma \beta = \oint_\gamma (-\lambda a\, dv)$$

is the puncture holonomy*. By Stokes it equals the bulk curvature integral $\int_\Sigma d\alpha$ over the punctured region Σ bounded by γ, so it is the analytic residue of the puncture — the same surface potential whose loop integral around a seam circle is the seam-circle holonomy of Proposition 9.7. For the two mechanisms of Theorem 8.9, this holonomy is the analytic companion of the two degenerations: the non-C^1 assignment (type 1) and the conformal-background critical point (type 2).*

Remark 8.12 (The puncture holonomy needs the declared data). As in Remark 9.8, the surface potential β is defined only where the arithmetic data (a, v) are declared on the punctured regular part; the holonomy of Proposition 8.11 is meaningful precisely under that declaration. Whether the holonomy converges to a well-defined residue as γ shrinks to p, and how that residue classifies the two puncture mechanisms, is open (part of OQ-096.3).

8.8 Smooth parameter families and their total zero sets

Let $I \subset \mathbb{R}$ be an open interval, let M be a smooth m-manifold without boundary, and let

$$A : M \times I \longrightarrow \mathbb{R}$$

be smooth. Write $a_t(p) = A(p, t)$ and

$$\mathcal{Z} = A^{-1}(0) = \{(p, t) \in M \times I : a_t(p) = 0\}.$$

Proposition 8.13 (Regular total zero set)**.** *Assume that*

$$d_p a_t \neq 0 \qquad \text{whenever } a_t(p) = 0. \tag{22}$$

Then $\mathcal{Z}$ is a smooth embedded codimension-one submanifold of $M \times I$, and the restriction

$$\pi : \mathcal{Z} \longrightarrow I, \qquad \pi(p,t) = t, \tag{23}$$

is a submersion. In particular, if $\dim M = 2$*, then $\mathcal{Z}$ is a smooth surface.*

Proof. At a point $(p,t) \in \mathcal{Z}$, the total differential is

$$dA_{(p,t)}(v,\tau) = d_p a_t(v) + d_t A_{(p,t)}(\tau).$$

Hypothesis (22) implies that $dA_{(p,t)} \neq 0$, so 0 is a regular value of A. The regular-value theorem proves that $\mathcal{Z}$ is a codimension-one embedded submanifold.

It remains to prove the stronger assertion about the projection. Given an arbitrary $\tau \in T_t I$, put $c = d_t A_{(p,t)}(\tau)$, and choose $w \in T_p M$ with $d_p a_t(w) = 1$, which is possible because $d_p a_t \neq 0$. Set

$$v = -c\, w.$$

Then $dA_{(p,t)}(v,\tau) = 0$, so $(v,\tau) \in T_{(p,t)}\mathcal{Z}$, while $d\pi_{(p,t)}(v,\tau) = \tau$. Hence $d\pi$ is surjective at every point of $\mathcal{Z}$, proving that π is a submersion. □

Remark 8.14 (Properness warning)*.* Proposition 8.13 gives pointwise product charts for the projection, but it does not make $\pi : \mathcal{Z} \to I$ a globally trivial family, or even a fiber bundle over an entire parameter neighborhood when noncompact fibers are present. A standard sufficient hypothesis for local triviality as a smooth fiber bundle is that the projection onto the relevant parameter interval be a proper surjective submersion (with the usual additional care if boundaries are introduced). Without properness, components may escape to or arrive from infinity. Accordingly, $\mathcal{Z}$ is called the *total zero set* here, not a zero tube.

No multi-zero model is included in this foundational chapter. Constructing and classifying such models requires a complete audit of the domain, metric, assignment, singular set, flow equation, and exact zero topology; that problem belongs to the singular-zero theory developed after Paper I.

9 Theory of Caps and Interface Disks

This chapter gives the "hole" and the "cap" a finite geometric carrier. A decorated directed acyclic network is thickened to a three-dimensional body whose side surface carries the holes and whose interior carries the interface disks; cutting the body along a disk exposes two caps.

9.1 The decorated ball-and-tube thickening

Definition 9.1 (Decorated ball-and-tube thickening)**.** Let G be a finite directed acyclic network with ordered typed external inputs and outputs, ordered input/output slots and deterministic partial functions at each gate, immutable sources, distinct occurrences, explicit copy/discard, and binding records. Forget the direction to obtain the underlying multigraph $|G|$. Choose, for each gate v, an oriented 3-ball B_v with disjoint parametrized port disks on its boundary, one per slot; for each internal wire e choose a tube $T_e = D^2 \times [0,1]$ and glue its end disks to the matching port disks, orientation-compatibly. The resulting body is H_G; the external port disks are E_G, and the observed side surface is

$$S_G = \partial H_G \setminus \bigcup_{D \in E_G} \operatorname{int}(D).$$

This is a declared minimal thickening rule, not an intrinsic geometry forced by the operation arity; the bare H_G or S_G carries no program identity.

Proposition 9.2 (Local pants and no-hidden-handle genus)**.** *1. A gate with total port count three has its sphere minus the three open port disks a* pair of pants*; total port count two gives a cylinder, one gives a disk. The* $2 \to 1$ *and* $1 \to 2$ *decorations are distinct.*

2. For a connected G*,* H_G *is a handlebody with*

$$g(H_G) = \beta_1(|G|), \qquad \chi(S_G) = 2 - 2\beta_1(|G|) - b,$$

where $b = |E_G|$ *is the number of external ports.*

Proof. (1) is the classification of surfaces with boundary. (2) Assemble along a spanning tree of $|G|$: each gate ball is a 0-handle, each tree edge glues two balls into one, and each remaining internal edge adds a 1-handle; hence the genus is the number of independent cycles. The closed boundary of a genus-g handlebody has Euler characteristic $2 - 2g$, and deleting b open port disks lowers it by b. □

9.2 Interface disks and the hole–disk complementarity

Proposition 9.3 (Pants gluing along an interface disk)**.** *Let two three-port gate balls be connected by one output–input tube. At the mid-section take the disk* $D_e = D^2 \times \{1/2\}$ *and its boundary circle* $C_e = \partial D_e$*. Cutting along* D_e *and absorbing the half-tubes into the two collars, the two bodies are* 3*-balls and the side surfaces are pants; regluing along* D_e *is, on the side surfaces, gluing along* C_e*:*

$$B_1 \cup_{D_e} B_2 \cong B^3, \qquad S_G \cong P_1 \cup_{C_e} P_2, \qquad \partial D_e = C_e.$$

Proof. Two balls joined by one 1-handle is again a ball (the underlying graph is a tree); the interior disk is not part of the exposed boundary, and its circle becomes the seam of the remaining boundary, a four-holed sphere with $\chi = -1 - 1 = -2$. □

Theorem 9.4 (Hole–disk complementarity)**.** *In the four-holed side surface* S_G *and the thickened body* H_G *of the preceding proposition:*

1. ***Surface non-fillability (hole).*** $[C_e] \neq 0 \in H_1(S_G; \mathbb{F}_2)$*:* C_e *bounds no disk inside* S_G*.*
2. ***Body fillability (disk/cap).*** *Under the inclusion* $i : S_G \hookrightarrow H_G$*,* $i_*[C_e] = 0 \in H_1(H_G; \mathbb{F}_2)$*, witnessed by the explicit disk* $D_e \subset H_G$ *with* $\partial D_e = C_e$*.*
3. ***Cutting exposes caps.*** *Cutting* H_G *along* D_e *splits the single internal disk into two copies* D_e^+, D_e^- *with opposite boundary orientations, each capping one pants opening.*

Hence the surface gives the hole its meaning and the body gives the disk its meaning, the two being linked by the oriented common edge $C_e = \partial D_e$*.*

Proof. (1) and (2) are the mod-2 homology computation recorded and verified in Appendix D: the side-surface boundary matrix has rank 110, rising to 111 on adjoining C_e, so C_e is not a boundary in S_G, while $C_e = \partial D_e$ is a boundary in H_G. (3) is the cut description of Proposition 9.3. □

Remark 9.5 (Down-set frontier disks). For a down-set $U \in \mathcal{O}(P)$ of the dependency poset of Theorem 2.17, cutting every internal tube that points from U to $P \setminus U$ yields the family of interface disks $\{D_e\}_{e \in \partial^+ U}$ representing the computational frontier.

9.3 The seam-circle holonomy

The hole–disk complementarity of Theorem 9.4 is topological. It has an analytic counterpart: the seam circle carries a holonomy of the surface potential, which is the boundary reading of the bulk curvature of Section 4.

Definition 9.6 (Surface potential)**.** Let the arithmetic data (a, v) be declared along the side surface S_G, with a the assignment and v the multiplicative coordinate. The *surface potential* is the one-form

$$\beta := (-\lambda a\, dv)\big|_{S_G},$$

the restriction to S_G of the primitive $-\lambda a\, dv$ whose exterior derivative is the contact curvature $d\alpha = -\lambda\, da \wedge dv$.

Proposition 9.7 (Seam-circle holonomy)**.** *For the seam circle $C_e = \partial D_e$ of Proposition 9.3, the loop integral*

$$\oint_{C_e} \beta = \oint_{C_e} (-\lambda a\, dv)$$

is the seam-circle holonomy*. By the holographic Stokes theorem of Proposition 5.5, this holonomy is the boundary reading of the bulk curvature $\int_\Sigma d\alpha$ over the disk $\Sigma = D_e$ that fills C_e; hence the same circle that is topologically non-fillable in S_G ($[C_e] \neq 0$) is analytically the carrier of a holonomy that the filling disk realizes.*

Proof. The first assertion is the definition. The second is Proposition 5.5 applied to the filling disk $\Sigma = D_e$ with boundary $\partial\Sigma = C_e$: the exactness of $d\alpha$ makes the bulk area integral equal to the boundary holonomy. The juxtaposition with the topological non-fillability is Theorem 9.4. □

Remark 9.8 (The analytic thickening is a declared structure). The surface potential β is well-defined only after the arithmetic data (a, v) are declared on S_G; the ball-and-tube thickening of Definition 9.1 is purely topological and does not by itself carry a, v. Extending the AES data from the bulk to the thickened body and restricting to S_G is the *analytic thickening*, a declared structure not yet constructed in general; the holonomy of Proposition 9.7 is meaningful precisely under that declaration. This is the analytic counterpart of the observation that a bare S_G carries no program identity.

9.4 Interface state, substitution, and the spectrum

Proposition 9.9 (Polynomial interface spectrum)**.** *In the rational-polynomial fragment, with $Q_+ : z = x + y$ and $P_\times : w = zt$ glued along the interface disk D_z, the substitution is the algebra homomorphism*

$$s^* : \mathbb{Q}[z] \to \mathbb{Q}[x, y], \qquad z \mapsto x + y,$$

and evaluation characters satisfy $\chi_{r,s} \circ s^* = \chi_{r+s}$. The "spectrum value" on the interface is the rational point $\chi_c(z) = c$ of $\operatorname{Spec} \mathbb{Q}[z]$*, and the relation gluing is the tensor product (geometric fiber product)*

$$\mathbb{Q}[x, y, z]/(z - x - y) \otimes_{\mathbb{Q}[z]} \mathbb{Q}[z, t, w]/(w - zt) \cong \mathbb{Q}[x, y, t, w]/(w - (x + y)t).$$

The spectrum $\operatorname{Spec} A_z$ *is a state-parameter space, not the geometric seam circle C_z and not the Stone order spectrum of Theorem 2.17.*

Proof. Direct substitution and elimination of z; the choice of this model as the AEG interface-spectrum semantics is a structural proposal, the displayed algebra itself being a proved identity in the polynomial fragment. □

Definition 9.10 (Interface record with residual)**.** An interface disk record is the five-tuple

$$\mathcal{I}_e = (D_e, C_e; \sigma_e, r_e, \beta_e),$$

where σ_e is the interface state under the declared observation layer, r_e is a recoverable production-process residual (together with any tangential geometric residual), and β_e is the port binding credential. Numerical equality of σ_e does not identify two production histories ($2+3$ and $1+4$ both carry $\sigma_e = 5$).

9.5 The capping obstruction and certified sealing

Theorem 9.11 (Constant-boundary capping obstruction)**.** *Let D be a compact smooth disk, g a smooth Riemannian metric, $a \in C^\infty(D)$, and $\mu \neq 0$, $\lambda \in \mathbb{R}$. There is no data (g, a) satisfying $|da|_g^2 = \mu^2 + \lambda^2 a^2$ on $\operatorname{int} D$ and $a|_{\partial D} = c$ (constant).*

Proof. Since $\mu^2 + \lambda^2 a^2 \geq \mu^2 > 0$, one has $da \neq 0$ throughout the interior. If a is nonconstant, compactness gives an interior extremum of a, where $da = 0$, a contradiction; if a is constant, $da = 0$ everywhere, also a contradiction. □

Corollary 9.12 (Capping dichotomy)**.**

1. *An interface disk with numeric label $\sigma_e = c$ must not be realized as a compact regular AES disk with constant boundary assignment $a|_{C_e} = c$.*
2. *To cap a level circle geometrically one must admit an interior critical point or a non-C^1/metric singularity (the singular category of Section 8).*
3. *To glue across a seam C_e in the regular category the boundary assignment $a|_{C_e}$ must be nonconstant and both sides must satisfy the two-sided smooth collar contract of Paper 0 (`def:p0-collar-contract`).*

Definition 9.13 (Certified cap tuple)**.** A candidate *certified cap* is a six-tuple

$$\mathcal{C} = (D, C, \sigma, r, w; \mathcal{Q}),$$

where D, C are the declared fill and its edge, σ a state, r a residual, w a verification certificate against an explicit closure obligation, and $\mathcal{Q}$ the declared observation scope. Four closure judgments are kept distinct: (i) computational port substitution binding, (ii) topological disk-filling $\partial D = C$ or conformal cusp filling, (iii) regular AES collar descent or essential-singularity removal, and (iv) the Adva `Seal { body }` witness closure (additive residual zero, multiplicative residual unit, with the non-vanishing domain obligations retained). A substitution success does not imply `Seal`, and $A = 0, M = 1$ does not by itself produce an AES cap.

10 Conclusions and Research Interfaces

This paper has established the syntactic and projective foundation of the programme. The organizing principle is that an expression should not be identified prematurely with its endpoint value. An unmarked sequential tree requires a choice of accumulator; the resulting fully marked planar tree is equivalent to a bounded marked history. From that history, operator evaluation and the charge

shadow form separate branches, while an initial value turns an operator into an endpoint. The non-bijective forgetful maps of (1) lose information, and the structures proved here are the ones that survive at the syntactic and operator levels.

10.1 Results proved here

The first part of the paper classified binary expression trees with a unique legal internal evaluation order and represented them by marked spinal histories. Operand-slot data made it possible to distinguish planar mirror, temporal reversal, and path inverse. These operations act at different levels and need not induce the same operator comparison.

For nondegenerate bilateral one-hole contexts, projective continuation produced Möbius transformations. The elementary contexts generate $\mathrm{PGL}_2(K)$ over an arbitrary field, quoting the generation theorem of Paper 0 and reading it through the history language, while ordinary addition and nonzero scaling form the affine, or Borel, sector stabilizing the chosen point at infinity. This placement is important: projective semantics extends the operator domain, but it does not turn a pole into an admissible ordinary arithmetic operation.

The smaller alphabet consisting of translation by $\pm\sqrt{2}$ and negative inversion realizes the $q = 4$ Hecke group inside this projective semantics. Its matrix relation $(JT)^4 = 1$ gives an explicit pair of different marked histories with the same operator. The subsequent factorization through cell stabilizers separates four levels that are needed downstream: history, operator, cell coset, and endpoint. In particular, arithmetic words are not identified one-to-one with tiles in a chosen Hecke cellulation.

The order-topological part of the paper extended the history layer to branching trees. The down-set information order gives a dcpo of finite and infinite computation DAGs, and the partial evaluations of a fixed tree form a finite distributive lattice whose maximal chains are the spinal histories. On the value lattice $[0, \infty]$, the elementary contexts realize the Scott–Lawson structure: the affine semigroup is the one-sided Scott carrier and the reciprocal involution is the step to the two-sided Lawson topology, mirroring the passage to $\mathrm{PGL}_2(\mathbb{R})$. The projective evaluation map has no Scott-continuous extension to the infinite histories, its only candidate limit being the continued fraction, which fixes the finite-history scope.

The analytic part of the paper established three pillars on this layer. The differential pillar identified the order defect of the minimal branching DAG with the contact curvature $[D_u, D_v] = \mu\lambda\,\partial_a$. The integral pillar expressed the order defect as a group-valued 1-cocycle on the linear-extension graph, closed by the Jacobi identity and by the holographic Stokes theorem. The series pillar expanded the history evaluation into the free Lie algebra as a Magnus series, whose arity-k brackets are carried by the $(k-1)$-cells of the down-set lattice and whose abelianization is the difference substitution.

The hole–cap part of the paper kept five notions of hole and four of boundary apart, thickened a decorated network to a ball-and-tube body exhibiting the hole–disk complementarity, and proved the constant-boundary capping obstruction.

The geometric part of the paper developed the zero theory of regular arithmetic expression spaces, using the imported interface of Section 7. At a zero of a regular assignment one has $|\nabla a| = |\mu| \neq 0$, so the interior zero locus is a smooth codimension-one set and cannot have isolated points, crossings, or branch points. Under the additional assumptions that the metric is complete and the manifold is connected and boundaryless, unit-gradient rectification gives the diffeomorphism

$$M \cong Z(a) \times \mathbb{R},$$

hence the zero locus is nonempty and connected. This is not asserted for incomplete models or

manifolds with boundary, and it is not in general a Riemannian product decomposition. The splitting theorem is therefore a no-go statement for branching zero networks in the regular category, and the paper accordingly defines singular arithmetic expression spaces by a closed nowhere-dense singular set admitting no local regular extension. Two essential puncture mechanisms are contrasted, one isolated-zero model is verified in detail; a smooth parameter family with spatially regular zeros has a submersive total zero set, and the properness warning records exactly how far that statement may be used.

The proved logical spine may therefore be summarized as

$$\text{sequential tree} \longrightarrow \text{marked planar sequential tree} \longleftrightarrow \text{bounded marked history},$$
$$\text{branching tree} \longrightarrow \text{dependency poset } P \longrightarrow \text{down-set dcpo } \mathrm{Idl}(\mathrm{FinDAG}) \longrightarrow \text{Birkhoff lattice } \mathcal{O}(P),$$
$$\text{marked history} \longrightarrow \text{projective operator} \longrightarrow \text{ordinary endpoint or pole},$$
$$\mathrm{Aff}^{+} \xrightarrow{\neg} \mathrm{PGL}_2 \longleftrightarrow \sigma \xrightarrow{\neg} \lambda,$$
$$\mathrm{Hist}_4 \longrightarrow G_4 \longrightarrow G_4/H_{\mathcal{C}} \longrightarrow \text{Hecke cell},$$
$$\text{minimal DAG defect } T_{\mathrm{open}} \longrightarrow [D_u, D_v] = \mu\lambda\,\partial_a \longrightarrow \text{Magnus series } \Omega(\gamma),$$
$$\text{regular AES} \longrightarrow Z(a) \text{ smooth} \longrightarrow M \cong Z(a) \times \mathbb{R} \text{ when complete},$$
$$\text{hole–disk complementarity} \longrightarrow \text{capping obstruction},$$
$$\text{singular set} \longrightarrow Z_{\mathrm{reg}}(a),\ Z_{\mathrm{sing}}(a).$$

This is a foundation for later developments, not a claim that every arithmetic expression, singularity, or computational phenomenon is already captured by these levels.

10.2 What moved to Paper 0

The geometric development that earlier versions of this paper contained is now proved in Paper 0: the arithmetic motions of an arithmetic expression space, the basic hyperbolic model $\mathfrak{E}_0$ and the punctured model $\mathfrak{E}_1$, the continuous arithmetic motion and its rectification, the point-path and ripple-pencil readings of a chronological matrix product, the affine cocycles and the relative affine defect, the accumulative commutative space with its weighted torsion–Stokes theorem, the reading of torsion as tearing, and the contact structure with curvature $\mu\lambda$. Those results keep their labels, hypotheses, and statuses; this paper cites them and does not maintain a second version. The same rule now applies to the generation theorem of §3: its proof, and the constructive factorization it yields, stand in Paper 0's chapter on matrix and projective unification, and this paper keeps only the history-level statement, the affine-sector corollary, and the Hecke sublanguage that are not developed there. The migration is recorded in `governance/00b-paper-0-geometric-foundation-amendment.md`.

10.3 Interface to Paper II

Arithmetic Expression Geometry II: Hyperbolic Real Function Theory begins where the horizontal connection of Paper 0 stops. It develops additional metric and analytic data: a horizontal metric, compatible complex or almost-complex structures where appropriate, second-order operators, boundary-value constructions, and arithmetic holomorphicity. None of these structures follows uniquely from the contact form alone, and no result of that downstream function theory is used in the proofs of this paper [8]. The G_4 sublanguage supplies an additional, explicitly arithmetic test case: an automorphic or equivariant function used there must be derived from the group action and must keep its analytic descent conditions separate from ordinary arithmetic admissibility of a history at a real input. The geometry imported by Paper II — the regular-AES definition, the canonical frame, the basic model, and the contact model — is imported from Paper 0.

10.4 Interface to Paper III

Arithmetic Expression Geometry III: Singular Zero Geometry and Tubes studies the boundary of the regular-zero theorem. Its objects include singular and multiple zeros, parameter discriminants, regular and singular zero surfaces, and tube constructions. Braid, monodromy, and knot structures there require separate invariance and normalization arguments. The Hecke interface proved here offers a source-driven alternative to imposing a four-valent cellulation by hand: Paper III uses a normalized Hauptmodul to construct the corresponding singular zero dessin, then relates its sign cover to Hecke periodic-orbit knots [9]. Those analytic, singular, and three-dimensional results remain downstream results, not results of this paper. The complete regular splitting theorem shows that such a branching network cannot be the zero set of one connected, complete, boundaryless regular scalar AES; any singularity, boundary, or incompleteness used in the construction must therefore be stated explicitly. In particular, the existence of a tube does not by itself produce a knot invariant, and no such general invariant is asserted here.

10.5 Interface to Paper IV

Arithmetic Expression Geometry IV: Projective Condensation and Computational Complexity returns to the full projective process–result hierarchy. Its scope includes bivaluation, quotient and condensation constructions, representation complexity, and the relation between geometric and computational resources. The marked-history formalism and the affine embedding in projective semantics proved here supply an entry point, and the affine torsion that Paper IV compares against is the one proved in Paper 0. Paper IV proves model-relative quotient and residual results together with fixed-model resource calibrations, while general machine-robust complexity claims remain open and require explicit models of representation, input size, and computation.

10.6 Open foundational problems

Several questions remain at the level of the history layer.

1. Characterize the marked-history fibers of the downstream G_4 operator–zero-network realization and determine what arithmetic decoration, if any, survives its operator and knot quotients.

2. Produce fully verified singular multi-zero models and identify their stable local normal forms and discriminants; the single verified isolated-zero model is a starting point, not a classification.

3. Determine when the total zero set of a parameter family is proper — hence a locally trivial tube — rather than merely a submersive family, and state the minimal hypotheses under which that upgrade is possible.

4. Formulate, and then test, a quantitative relation between history complexity, representation complexity, and computational complexity, with an explicit state space, encoding, and cost model.

5. Determine which relations among marked histories are compatible with ordinary admissibility, as opposed to projective non-degeneracy alone; the Hecke order-four example shows that the two conditions are genuinely different.

These are research interfaces rather than consequences of the present paper. The stable conclusion is narrower and more concrete: sequential arithmetic histories possess projective operator semantics with a distinguished arithmetic sublanguage; their affine sector is the Borel sector of that semantics;

and the regular zero geometry of an arithmetic expression space is rigid in exactly the sense stated above, so that singular and branching phenomena require an explicit change of category.

Acknowledgements and provenance

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DeepSeek Harness, an artificial-intelligence research agent running the model `deepseek-v4-flash-vision-exp`, assisted with proof drafting, with cross-checking of the finite classifications and matrix identities stated here, with red-team reading of the claim statuses, and with the preparation of LaTeX source. An agent is not a natural person: it cannot hold copyright, cannot sign a transfer, and cannot correspond, so it appears on the title page as assistance and not as a coauthor. Contribution-level provenance, including the commits and model identities where they are known, is maintained in `AUTHORSHIP_AND_PROVENANCE.md`; the same document records the standing rule that machine cross-checking is not independent human review. The mathematical statuses used here, and the boundary between proved statements and structural proposals, are those registered in `governance/05-mathematical-status.md`.

A Composition, action, and domain conventions

This appendix collects the convention-sensitive choices used in Sections 2–3. The choices are not mathematical assertions, but every composition, sign, and coordinate formula depends on them. The affine-group conventions that belong to the cocycle and flow chapters of Paper 0 are collected in that paper's appendix.

A.1 Chronological words and functional composition

Convention A.1 (Order inside a history)**.** A history

$$\gamma = (g_1, \dots, g_n)$$

performs g_1 first and g_n last. Its functional evaluation is therefore

$$\mathrm{ev}(\gamma) = g_n \circ \cdots \circ g_1$$

when the g_i are viewed as maps.

Convention A.2 (Chronological concatenation)**.** The written product $\gamma\delta$ means first perform γ and then perform δ. Thus

$$\mathrm{ev}(\gamma\delta) = \mathrm{ev}(\delta) \circ \mathrm{ev}(\gamma).$$

The notation is chronological rather than functional: the earlier process is written on the left.

For example, if $g(z) = z + \mu$ and $h(z) = qz$, then the word $\gamma = (g, h)$ and the chronological product of the one-step histories gh both act as

$$h \circ g(z) = qz + q\mu.$$

The reverse chronological word hg acts as $g \circ h(z) = qz + \mu$. This two-step check fixes the sign of their endpoint difference:

$$\mathrm{ev}(gh)(z) - \mathrm{ev}(hg)(z) = \mu(q - 1).$$

A.2 Column-vector projective action

Convention A.3 (Matrix action)**.** Homogeneous coordinates are columns, and matrices act from the left:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} Au + Bv \\ Cu + Dv \end{pmatrix}.$$

On the affine chart $z = [z : 1]$, this gives

$$M \cdot z = \frac{Az + B}{Cz + D}.$$

If M_i represents g_i, the history $(g_1, \ldots, g_n)$ is represented by

$$M_n \cdots M_1.$$

Consequently, for chronological concatenation,

$$\rho(\gamma\delta) = \rho(\delta)\rho(\gamma). \tag{24}$$

This is the matrix version of Convention A.2. A row-vector right action would require a different product convention; it is not used in this paper.

Projective matrices are scalar classes. If M and aM differ by $a \in K^\times$, they represent the same element of $\mathrm{PGL}_2(K)$. Matrix identities used for projective maps may therefore be checked either exactly in $\mathrm{GL}_2(K)$ or up to one overall nonzero scalar, provided the choice is stated.

A.3 Ordinary and projective domains

The following table records the domain distinction for the contexts involving division. Here $c \in K$ and z is the evolving affine value.

Context	Ordinary rule	Ordinary condition	Projective status
$C^{(1)}_{/,c}[z] = z/c$	divide by fixed c	$c \neq 0$	invertible if $c \neq 0$
$C^{(2)}_{/,c}[z] = c/z$	divide by evolving z	$z \neq 0$	invertible if $c \neq 0$
$C^{(\varepsilon)}_{\times,c}[z] = cz$	multiply by c	none	invertible if $c \neq 0$

Two different exclusions appear in the table. The parameter condition $c \neq 0$ ensures that the context has an invertible projective matrix. The state condition $z \neq 0$ ensures that an ordinary second-slot division is defined at that step. Projectively, a non-degenerate c/z sends 0 to ∞; this is a chart transition, not an ordinary quotient in K.

For a composite history, ordinary admissibility is checked at every intermediate state. It is not enough that the final projective point lies in the affine chart. In particular, two projective inversions compose to the identity operator, but the corresponding ordinary two-step word is not admissible at zero.

A.4 Mirror, reversal, and inverse notation

For $\gamma = (g_1, \ldots, g_n)$, the three operations are summarized by

Operation	Resulting word	Required hypothesis
Mirror $m\gamma$	$(m(g_1), \dots, m(g_n))$	none
Temporal reversal $r\gamma$	$(g_n, \dots, g_1)$	none
Path inverse γ^{-1}	$(g_n^{-1}, \dots, g_1^{-1})$	every g_i invertible

Mirror exchanges operand slots and retains time. Temporal reversal changes time and retains the original contexts. Path inverse reverses time and inverts every operator. Accordingly,

$$\rho(\gamma^{-1}) = \rho(\gamma)^{-1},$$

whereas no analogous identity holds for $r\gamma$ in general.

A.5 Notation conversion summary

The table below lists older or potentially ambiguous descriptions and the notation used in Paper I.

Potentially ambiguous description	Paper I notation
evolving value in operand slot one	$C^{(1)}_{\omega,c}[z] = \omega(z, c)$
evolving value in operand slot two	$C^{(2)}_{\omega,c}[z] = \omega(c, z)$
chronological context word	$\gamma = (g_1, \dots, g_n)$, g_1 first
history product	$\gamma\delta$, first γ, then δ
operator evaluation	$\rho(\gamma)$
ordinary endpoint at x	$\nu_x(\gamma)$
operand-slot exchange	$m\gamma$
time-order reversal	$r\gamma$
inverse operator path	γ^{-1}

B Equality and elementary neutrality examples

The examples below separate several equality tests without developing a general relation or loop theory. In each case, histories are marked chronological words, operators are affine maps, and endpoint values are obtained only after choosing an input.

Two step families are used throughout, with the notation and the parameters of Paper 0:

$$\mathsf{A}_p(z) = z + p, \qquad \mathsf{M}_q(z) = e^q z, \qquad p, q \in \mathbb{R}.$$

Negative p and q are allowed, so the same notation covers inverse translations and inverse positive scalings. For a word γ of these steps we write $A_\gamma = \sum_i p_i$ and $M_\gamma = \sum_i q_i$ for its two totals. Paper 0 constructs the space of which the pair (A_γ, M_γ) is the endpoint, proves that this endpoint is the abelianization of the history group, and proves the theorems that measure the order defect of two histories with equal totals; none of those theorems is restated here. The totals are used below only as bookkeeping, to say exactly what a pair of totals retains and what it forgets.

B.1 Same value, different operators

Let

$$\gamma = (\mathsf{A}_1), \qquad \delta = (\mathsf{M}_{\log 2}).$$

At the input $x = 1$, both histories have endpoint value 2. Their induced operators are nevertheless

$$\nu_x(\gamma) = x + 1, \qquad \nu_x(\delta) = 2x,$$

which are unequal. Equality at one input therefore does not imply operator equality.

B.2 Same operator, different histories

The marked histories

$$\gamma = (\mathsf{A}_1, \mathsf{A}_2), \qquad \delta = (\mathsf{A}_2, \mathsf{A}_1)$$

are different chronological words, but both induce $x \mapsto x + 3$. They also have the same totals, $A_\gamma = A_\delta = 3$ and $M_\gamma = M_\delta = 0$. Thus operator equality does not recover a marked history, even when the totals are retained.

B.3 Neutral endpoint, nontrivial history

For $p \neq 0$, the nonempty history

$$\ell = (\mathsf{A}_p, \mathsf{A}_{-p})$$

induces the identity operator and hence has zero endpoint defect for every input. It is nevertheless not the empty marked history. In particular, operator neutrality does not imply literal or historical equality.

A slightly different phenomenon occurs for

$$\gamma = (\mathsf{A}_p, \mathsf{M}_q, \mathsf{A}_{-p}, \mathsf{M}_{-q}), \qquad p, q \neq 0.$$

Its totals vanish, $A_\gamma = 0$ and $M_\gamma = 0$, but it induces the translation

$$z \longmapsto z + p(1 - e^{-q}),$$

which is not the identity. Thus vanishing totals do not imply operator neutrality, even though the individual steps are all non-degenerate and ordinarily admissible on an open set.

These examples exhibit the hierarchy needed in the main text:

$$\text{marked history} \longrightarrow \text{operator} \longrightarrow \text{endpoint value},$$

with the two totals providing an additional, order-forgetting projection. No converse implication between adjacent levels holds in general.

Two further separations belong to Paper 0 and are not repeated here: histories with equal totals but nonzero relative defect, and the distinction between the open two-step defect and the closed-loop drift of the same word. They are proved there as the relative torsion of two charge-compatible histories and as the finite comparison of the two orders, in the chapters on the accumulative commutative space and on the contact model, and their computations are collected in Paper 0's appendices on affine cocycles and on the ACS and contact calculations.

C Symbolic verification of the Magnus series and PBW abelianization

This appendix records the symbolic verification of the Magnus-series low-order terms of Definition 6.1 and the PBW abelianization of Theorem 6.5.

C.1 The order-three free nilpotent matrix model

Work in the free order-three nilpotent Lie algebra $\mathfrak{n}_4$ of strictly upper-triangular 4×4 matrices, realized by the elementary matrices

$$X = E_{12}, \qquad Y = E_{23}, \qquad Z = E_{34},$$

whose only nonzero products are $XY = E_{13}$, $YZ = E_{24}$, $XYZ = E_{14}$; all brackets of depth four or more vanish. The relevant brackets are

$$[X, Y] = E_{13}, \qquad [Y, Z] = E_{24}, \qquad [X, [Y, Z]] = [Z, [X, Y]] = E_{14}, \qquad [Y, [Z, X]] = 0.$$

C.2 Symbolic expansion to order three

For scalar parameters h, k, l, form the truncated exponentials

$$e^{hX} = I + hX + \tfrac{h^2}{2}X^2 + \tfrac{h^3}{6}X^3,$$

and similarly for Y, Z; here $X^2 = 0$ in the present model, so the exponential is a polynomial. The chronological product $e^{lZ}e^{kY}e^{hX}$ has formal logarithm obtained from the series $\log(I + T) = T - \frac{1}{2}T^2 + \frac{1}{3}T^3$, with $T = e^{lZ}e^{kY}e^{hX} - I$. The nonzero upper-triangular entries are

$$\Omega_{12} = h, \qquad \Omega_{23} = k, \qquad \Omega_{34} = l, \qquad \Omega_{13} = -\tfrac{1}{2}hk, \qquad \Omega_{24} = -\tfrac{1}{2}kl, \qquad \Omega_{14} = \tfrac{1}{3}hkl.$$

These are the coefficients claimed in Proposition 6.2: the order-one terms h, k, l are the arity-1 charges; the order-two terms $-\frac{1}{2}hk = -\frac{1}{2}[X, Y]$, $-\frac{1}{2}kl = -\frac{1}{2}[Y, Z]$ are the arity-2 commutators (equivalently $\frac{1}{2}hk[Y, X]$, $\frac{1}{2}kl[Z, Y]$, matching the sum $\frac{1}{2}\sum_{i<j}[X_j, X_i]$ of Definition 6.1); and the order-three term $\frac{1}{3}hkl$ is the arity-3 bracket $[X, [Y, Z]]$. The repeated-generator self-nesting terms $[X, [X, Y]]$ and $[Y, [X, Y]]$ vanish in $\mathfrak{n}_4$ because $X^2 = Y^2 = 0$; they are nonzero only in a step-3 model with nonzero squares, which is not needed for the low-order arity correspondence.

C.3 PBW abelianization check

Setting all brackets to zero abelianizes $\mathfrak{L}$: the bracket monomials $[X, Y] \mapsto 0$, $[X, [Y, Z]] \mapsto 0$, and the Magnus series degenerates to the abelian sum

$$\Omega(\gamma) \mapsto hX + kY + lZ,$$

whose coefficient vector (h, k, l) is the cumulative sum of the increments — the abelian form of the difference substitution of Theorem 6.5. Under the PBW filtration, the word ZYX (i.e. $Z \otimes Y \otimes X$) has associated graded the monomial XYZ in the symmetric algebra $S(\mathfrak{L})$, which is the commutative degeneration of the Lyndon word $[X, [Y, Z]]$ together with the abelian part.

D Finite cellulation and mod-2 homology verification for pants gluing

This appendix records the finite cellulation and mod-2 homology verification of Theorem 9.4 (V1 of OQ-096).

D.1 The V1 finite model

Two solid boxes $[0,3]^3$ and $[5,8]\times[0,3]^2$ model two three-port gates, joined by the connecting tube $[3,5]\times[1,2]^2$. The cellulation has 56 unit cubes, 336 tetrahedra, and 132 vertices. Every vertex link was checked for being a disk or a sphere (116 disks, 16 spheres), establishing the local manifoldness of the thickened body.

D.2 Surface and body homology data

The two pants side surfaces each have $(V,E,F)=(60,116,55)$ with three boundary circles; the glued side surface has $(V,E,F)=(116,228,110)$ with four boundary circles and Euler characteristic $\chi=-2$, matching Proposition 9.2 ($\beta_1=0$, $b=4$). The shared interface disk is one square; its four oriented edges form the seam circle C, with opposite orientation on the two sides.

The mod-2 side-surface boundary matrix has rank 110; adjoining the chain C raises the rank to 111, proving $[C]\neq 0\in H_1(S_G;\mathbb{F}_2)$. In the body, the explicit disk gives $\partial D=C$, so $i_*[C]=0\in H_1(H_G;\mathbb{F}_2)$.

D.3 Negative controls

Ten negative controls were executed and all rejected: a control is accepted only if it violates a declared condition (wrong orientation, a non-manifold link, a mismatched boundary count, or a spurious disk filling in the side surface). The substitution $z=x+y$, $w=zt$ at $(x,y,t)=(2,3,4)$ gives interface value 5 and output 20 by exact rational arithmetic, agreeing with the graph binding.

D.4 Status

This is a *computationally verified example* for a single fixed model. The construction and the checker are by the same author; it does not close the general composition, the AES-extension, or the native `Seal` bridge obligations (V2–V4 of OQ-096).

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